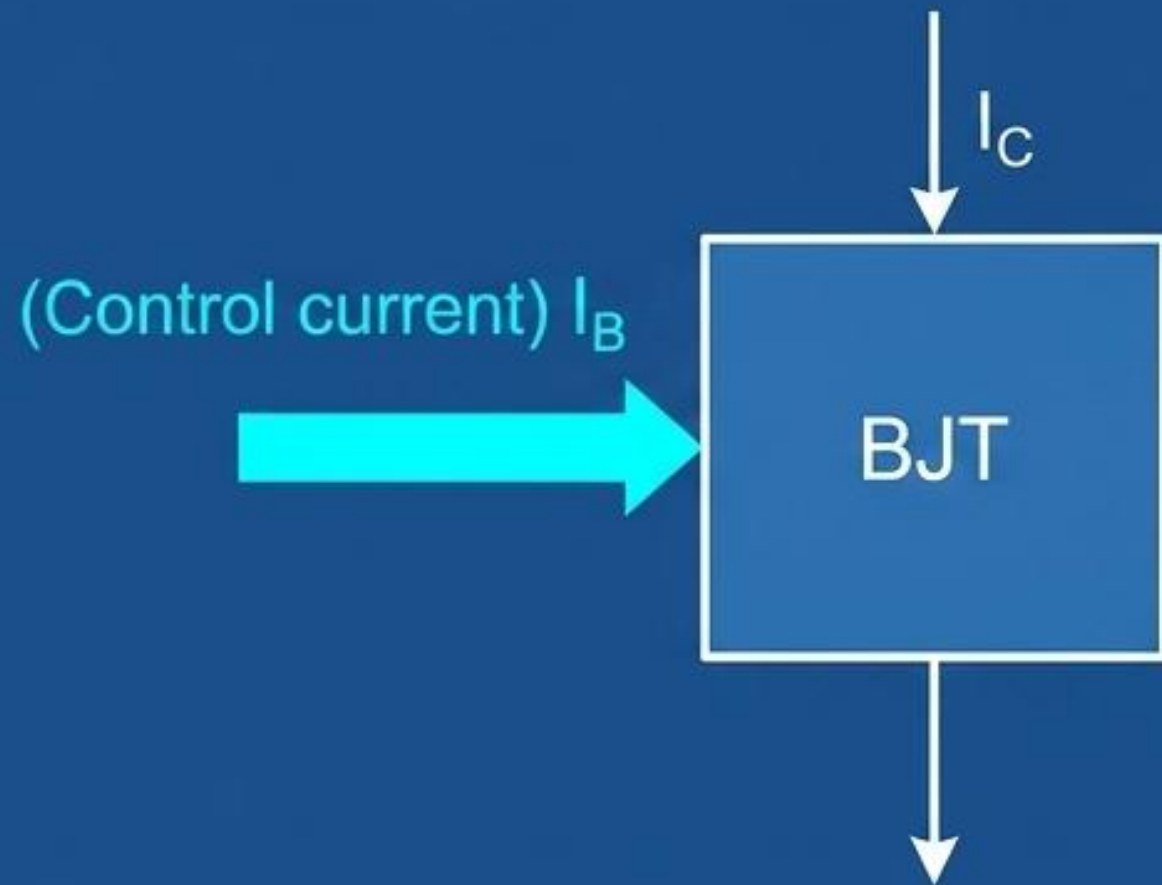


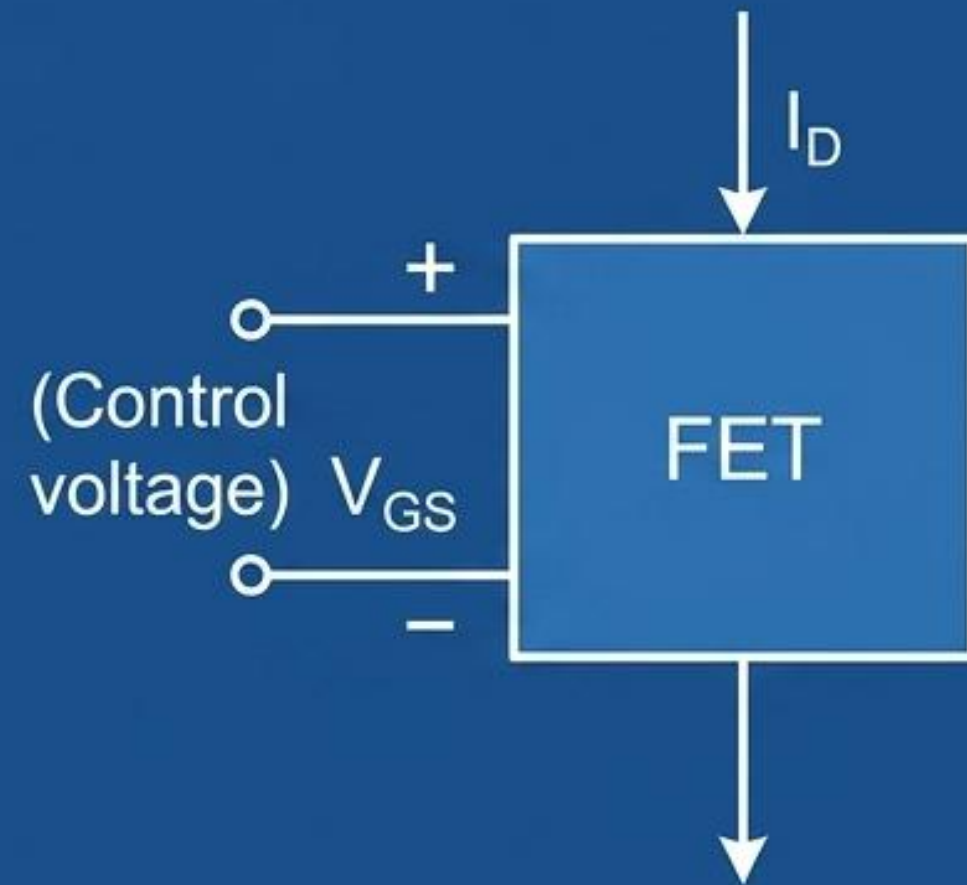
Field-Effect Transistors

Topic 6 (Chapter 6)

The Fundamental Shift: Current Control vs. Voltage Control



BJT (Bipolar Junction Transistor):
A current-controlled device.
The base current dictates operation.



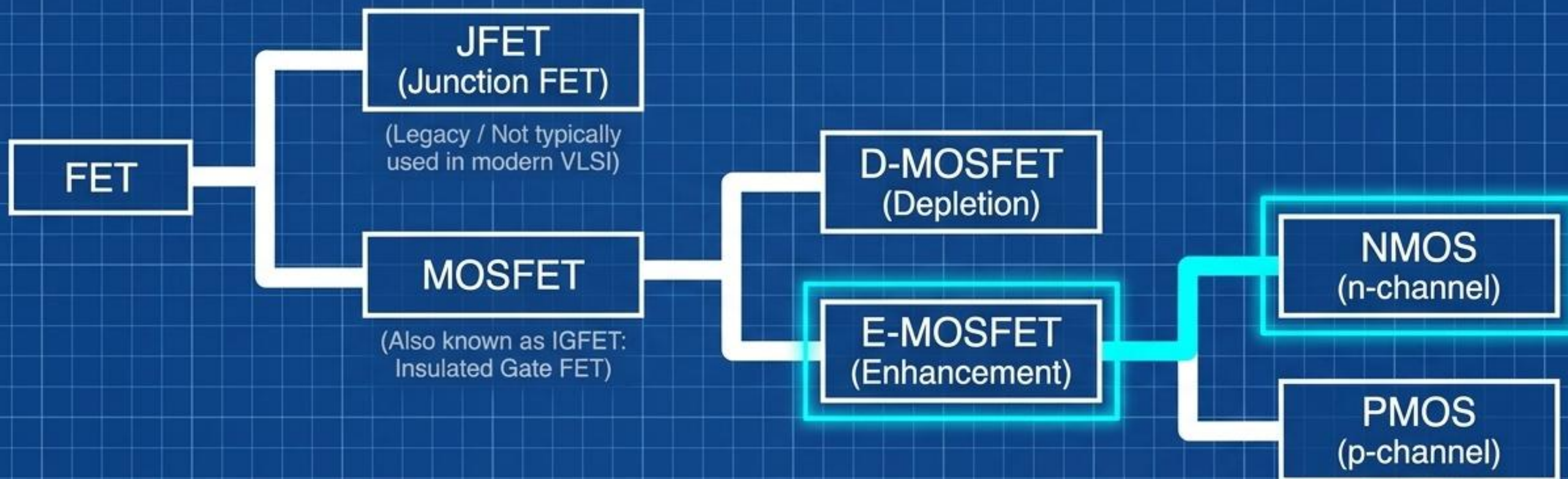
FET (Field-Effect Transistor):
A voltage-controlled device.
An electric field dictates operation.

Comparative Matrix: Evaluating Transistor Architectures

Dimension	BJT	FET
Polarity	Bipolar	Unipolar
Control Mechanism	Current-controlled	Voltage-controlled
Input Impedance	Lower	Significantly Higher
Gain	Higher	Lower
Switching Speed	Slower	Faster
Form Factor	Larger	Smaller
Thermal Sensitivity	Highly sensitive	Less sensitive (Optimal for ICs)

Similarities: Both are 3-terminal transistors utilized universally as amplifiers and switching devices.

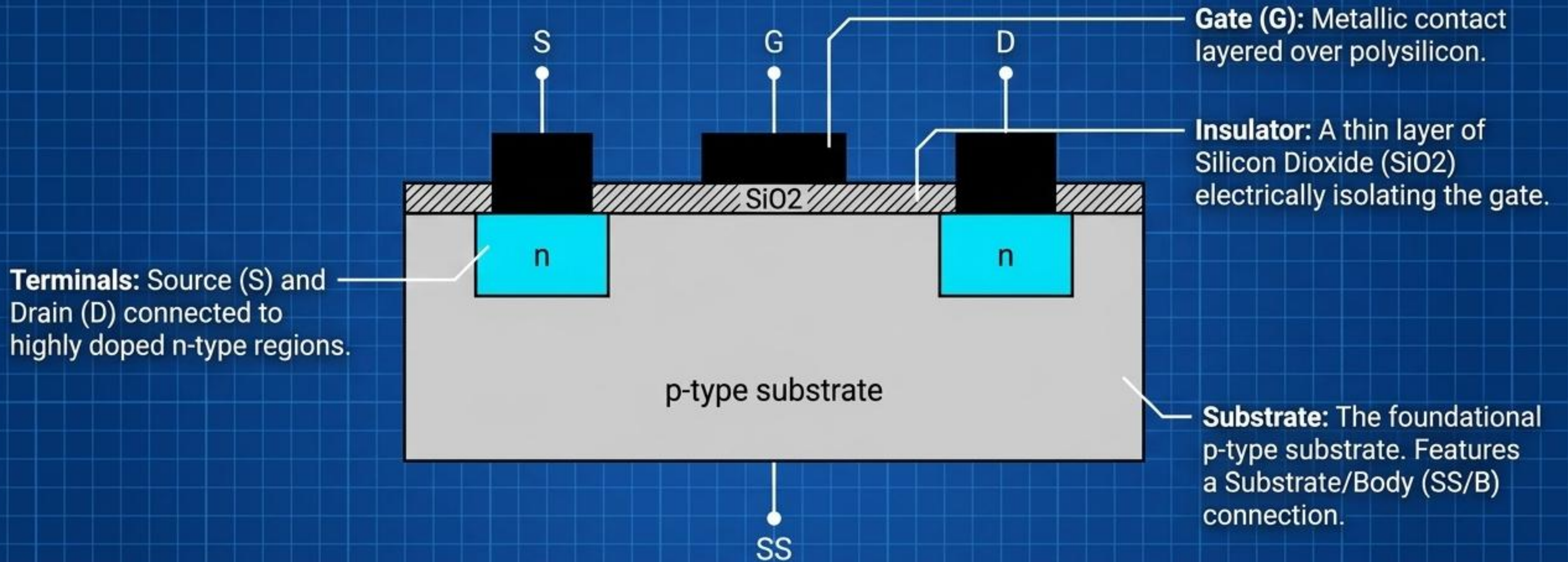
The Transistor Taxonomy: Locating the E-MOSFET



Nomenclature Note:

Despite the name (Metal-Oxide-Semiconductor), modern MOSFET gates actually use polysilicon instead of metal.

Anatomy of an NMOS: The Baseline State

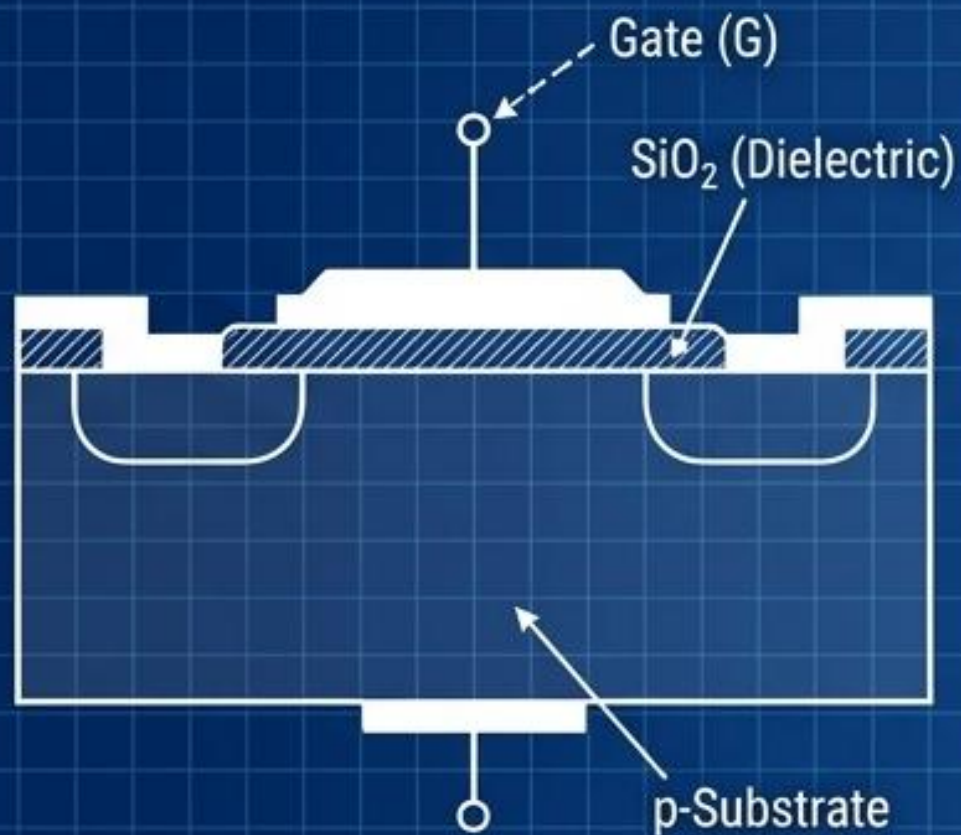


The Critical Gap:

Under no-bias conditions, there is strictly no channel connecting the Source and Drain. The device is fundamentally "Off" by physical design.

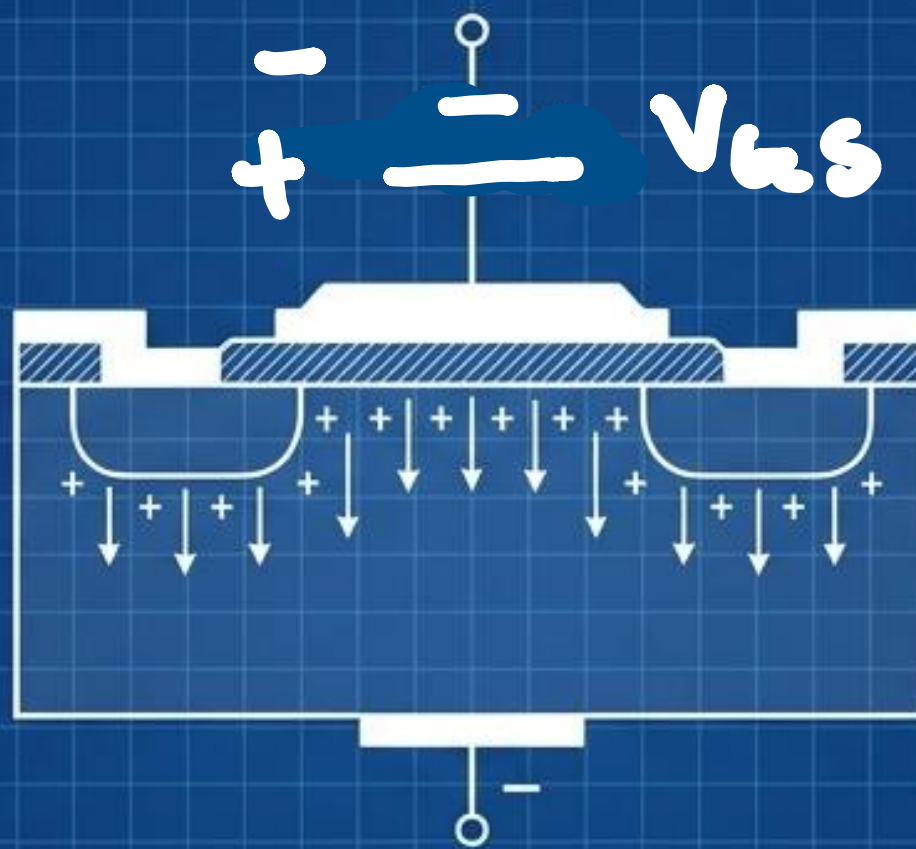
The Field Effect: Inducing a Conductive Channel

The Parallel-Plate Capacitor



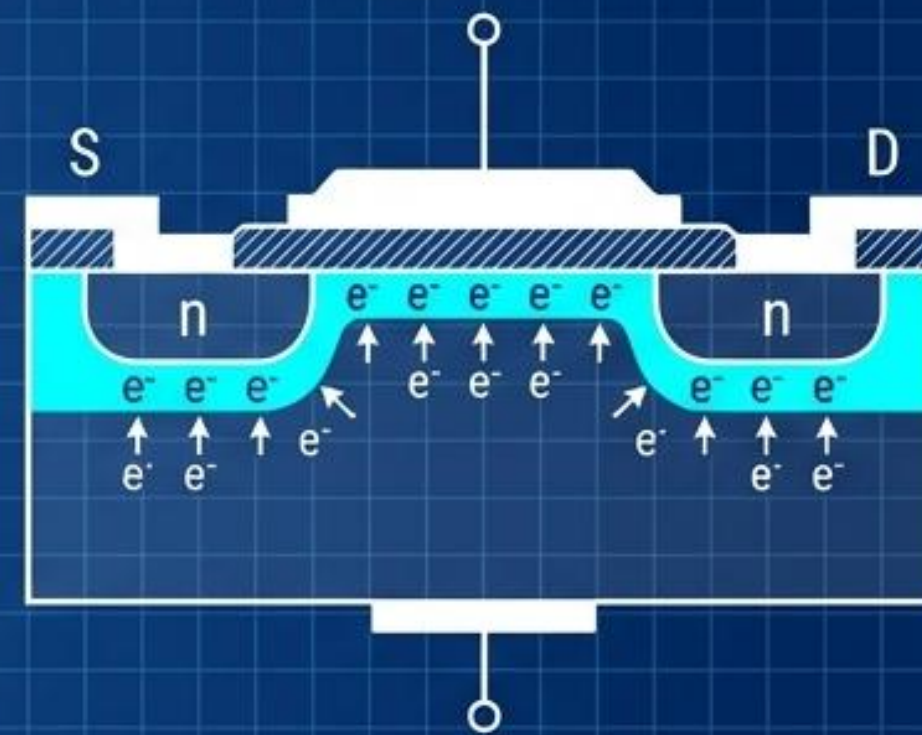
The Gate, dielectric, and Substrate act as a capacitor. Gate current (I_G) is strictly 0A.

Repelling Holes



A positive voltage (V_{GS}) applied to the Gate repels the positive holes downward, creating a depletion region.

Attracting Electrons

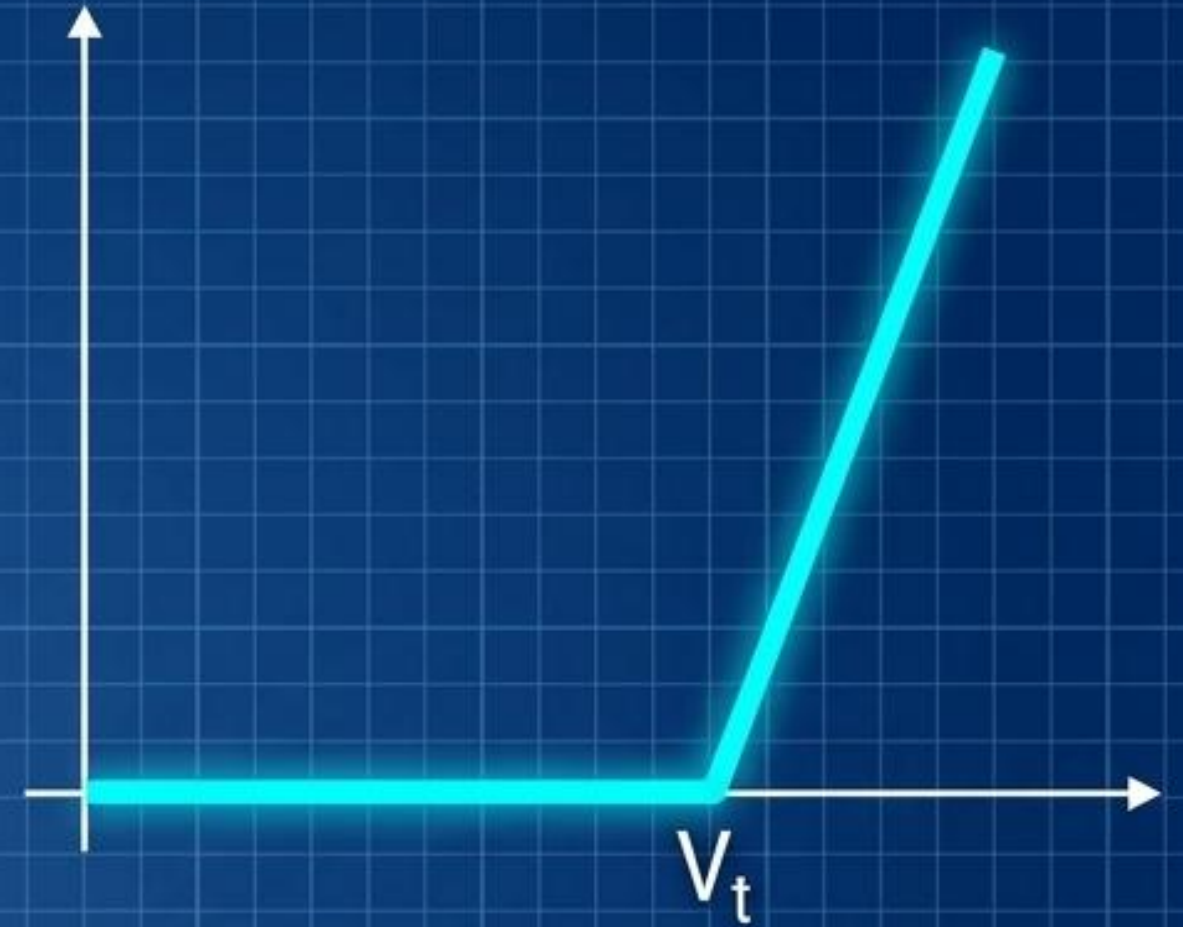


The positive field pulls minority carriers (electrons) to the surface. A continuous n-channel now bridges the Source and Drain.

Takeaway: The potential difference V_{DS} creates the current flow, but the electric field created by V_{GS} creates the channel itself.

The Turning Point: Threshold Voltage (V_t)

$$V_{GS} \geq V_t$$



Defining V_t :

The precise minimum value of V_{GS} at which a sufficient number of mobile electrons accumulate to form a conducting channel.

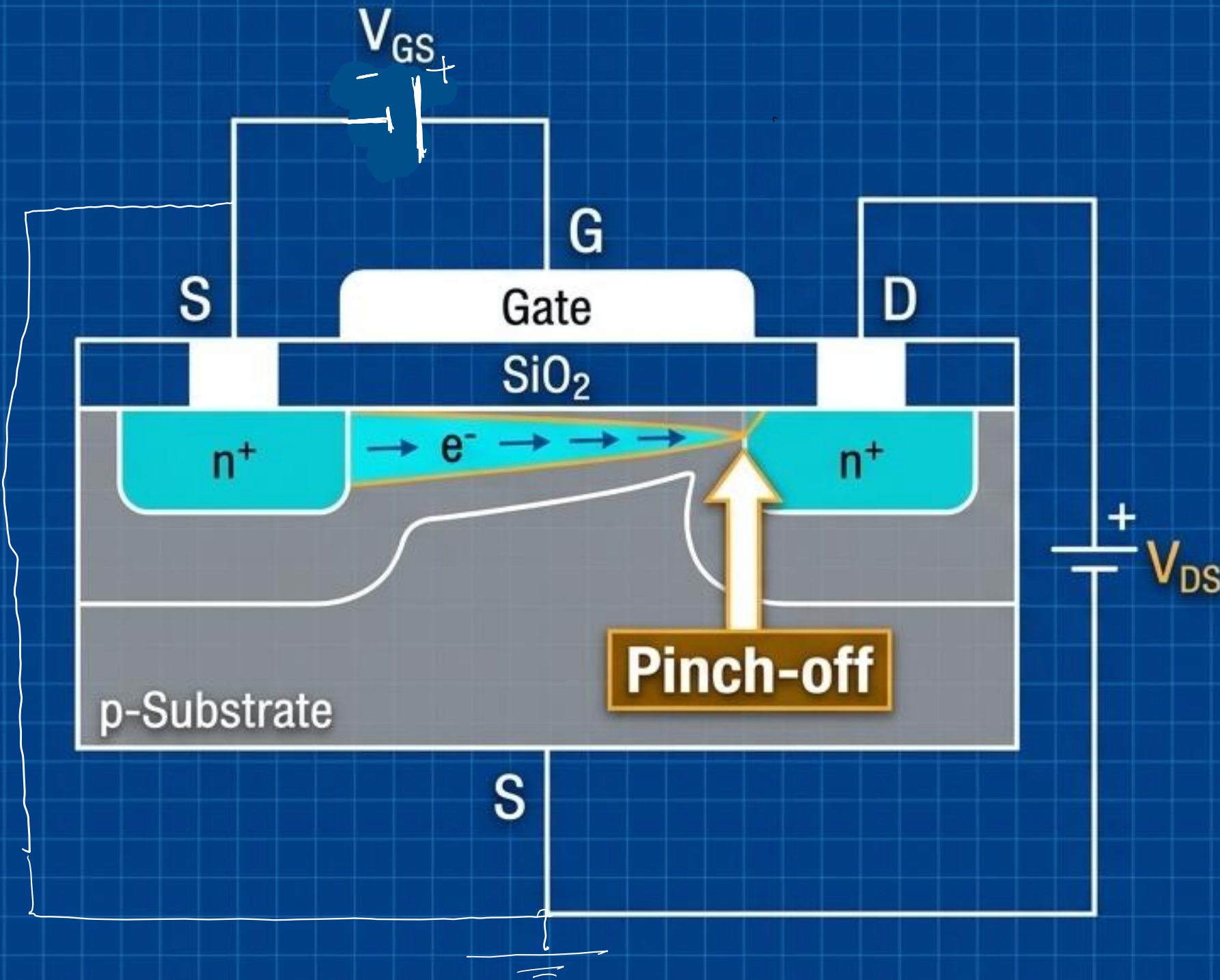
- It is a fixed hardware parameter.
- Controlled entirely during device fabrication.
- Typically ranges from 0.3V to 1.0V for modern NMOS.

Nomenclature: Why 'Enhancement-Type'?

For this MOSFET to conduct, a channel must first be induced from nothing.

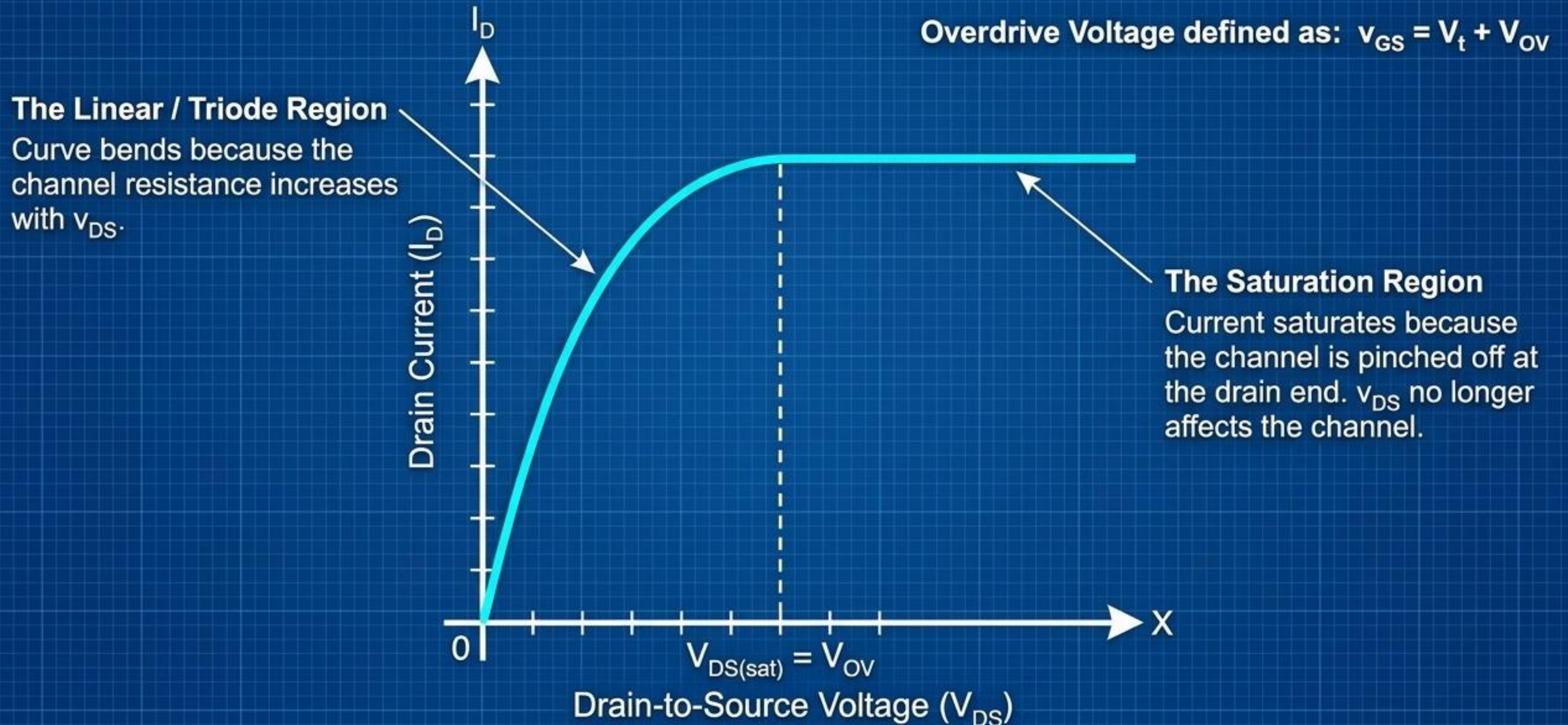
Increasing the voltage of V_{GS} above the threshold V_t physically "enhances" the density and depth of this conductive channel.

Channel Dynamics: The Pinch-Off Effect



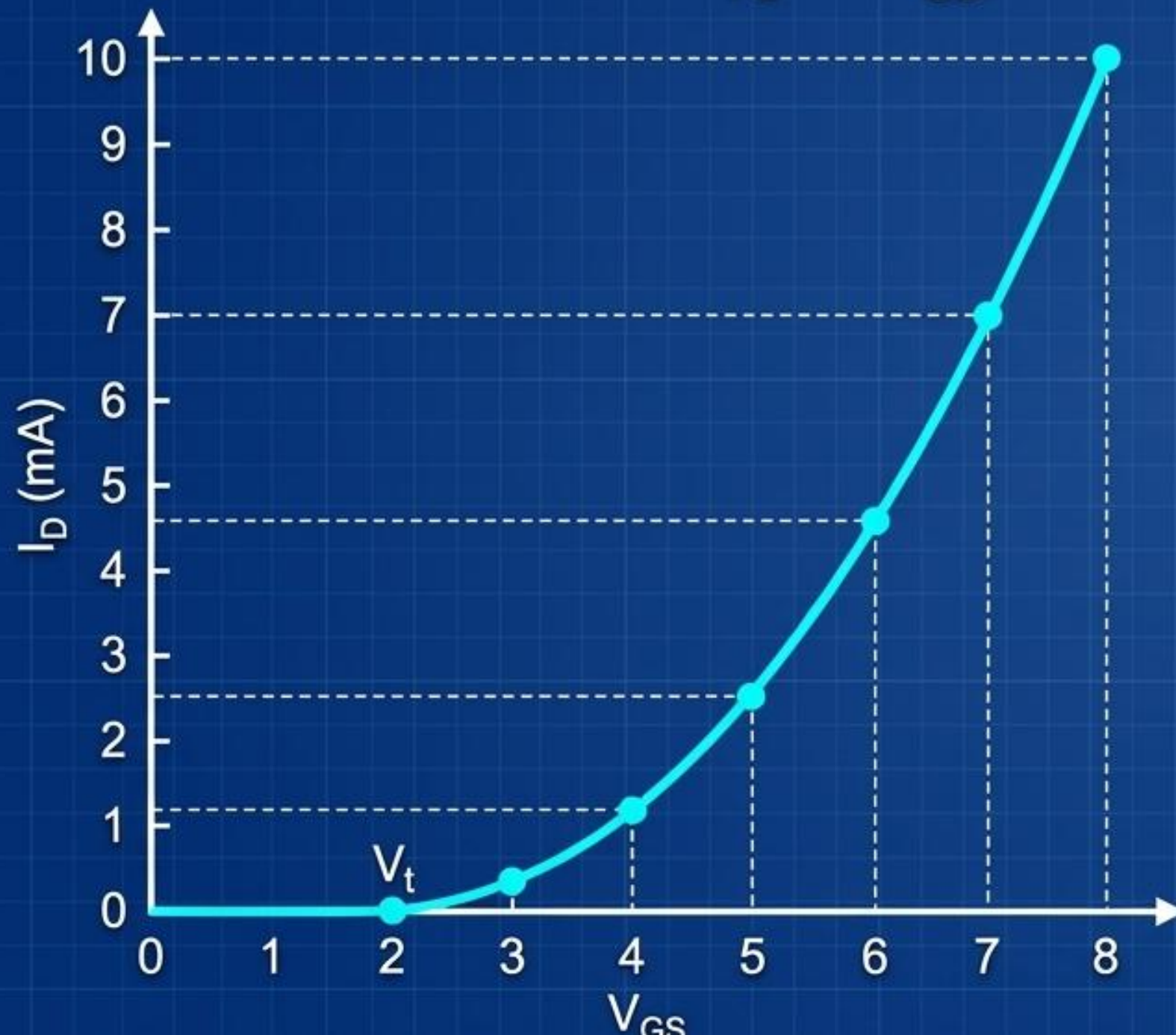
- Assume a fixed $V_{GS} > V_t$
The channel is successfully opened
- Increase the Drain Voltage (V_{DS})
Current begins to flow from Drain to Source
- The Voltage Potential Varies
The potential difference between the Gate and the channel changes along its physical length. Near the drain, the effective voltage is $V_{GD} = V_{GS} - V_{DS}$.
- The Tapering Effect
As V_{DS} increases, V_{GD} decreases. The channel width dynamically shrinks at the drain end until it eventually 'pinches off'.

The Characteristic Curve: Mapping Transistor States



The Transfer Curve: Defining Transconductance

Transfer Curve (I_D vs V_{GS})



$$I_D = k(V_{GS} - V_t)^2$$

Variable Breakdown:

- V_t : **Threshold voltage**. The minimum voltage to induce a channel.
- k : **Transconductance parameter (A/V^2)**. A constant tied exclusively to the device's physical construction and geometry.

Calculation Box: Solving for k

Determine k by utilizing values at a known operating point on the curve:

$$k = \frac{I_{D(ON)}}{(V_{GS(ON)} - V_t)^2}$$

Mathematical Modeling: The Three States of Operation

State 1:
OFF Region (Cutoff)

$$V_{GS} < V_t$$

(or $V_{GS} = 0V$)

$$I_D = 0A. \text{ (Note: } I_G \text{ is always } 0A\text{).}$$

State 2:
Linear / Triode Region

$$V_{GS} > V_t$$

AND

$$V_{DS} < (V_{GS} - V_t)$$

$$I_D = 2k \left[(V_{GS} - V_t)V_{DS} - \frac{1}{2}V_{DS}^2 \right]$$

(For very small V_{DS} , $I_D \approx 2k(V_{GS} - V_t)V_{DS}$)

State 3:
Saturation Region

$$V_{GS} > V_t$$

AND

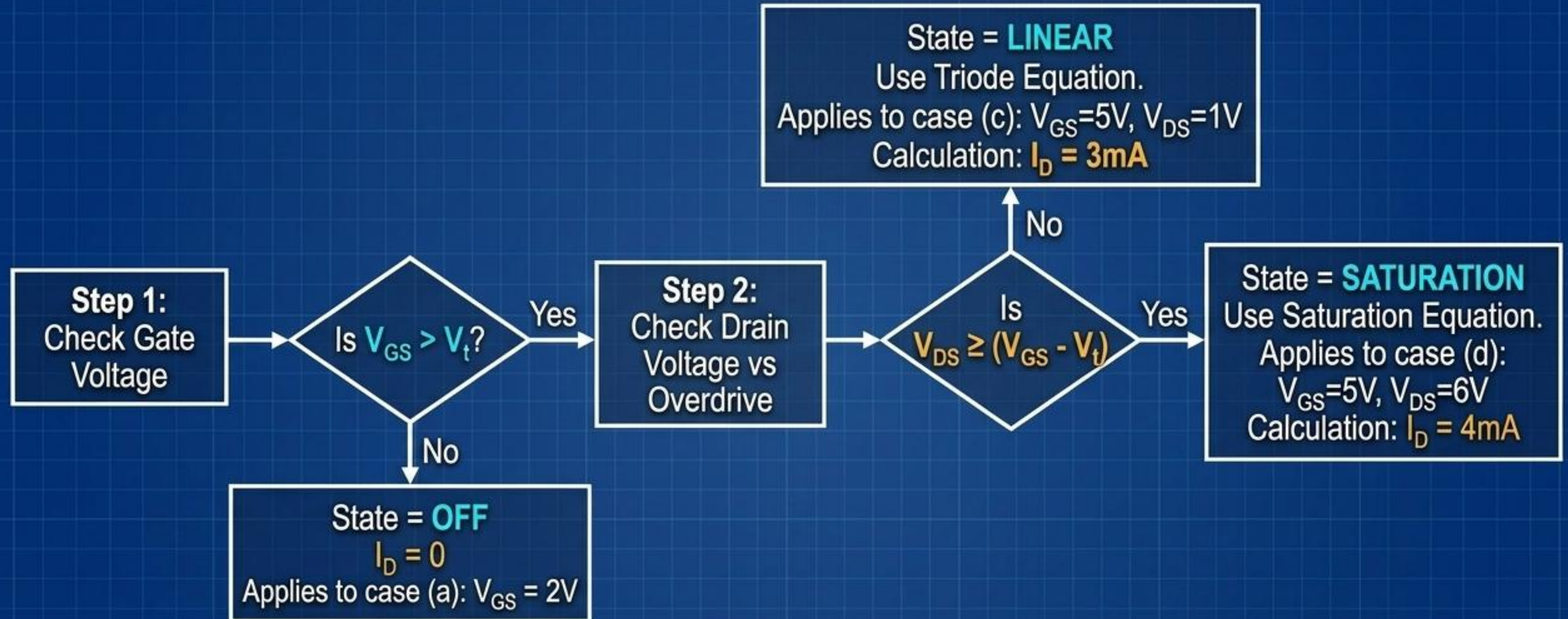
$$V_{DS} \geq (V_{GS} - V_t)$$

$$I_D = k(V_{GS} - V_t)^2$$

$(V_{GS} - V_t)$ is commonly referred to as the Overdrive Voltage (V_{OV}).

Diagnostic Flowchart: Solving for Current

Given: NMOS | $k = 1 \text{ mA/V}^2$ | $V_t = 3\text{V}$



Note: Gate current (I_G) is structurally 0A. Source current (I_S) equals Drain current (I_D).
If $V_{DS} = 0\text{V}$, the device is functionally OFF regardless of V_{GS} (case b).

EMOSFET CURRENT PROBLEM (Enhancement NMOS)

- For an enhancement NMOS, $k = 1 \text{ mA/V}^2$, $V_t = 3 \text{ V}$.
- For each case, identify the region of operation and find I_D , I_S and I_G . Write unit.

1. REGION OF OPERATION

- Cutoff (OFF):** $V_{GS} < V_t$
- Triode (Linear):**
 $V_{GS} > V_t$ and $V_{DS} < V_{GS} - V_t$
- Saturation:**
 $V_{GS} > V_t$ and $V_{DS} \geq V_{GS} - V_t$

2. CURRENT EQUATIONS

- Triode (Linear) Region:**

$$I_D = 2k \left[(V_{GS} - V_t)V_{DS} - \frac{1}{2}V_{DS}^2 \right]$$

- Saturation Region:**

$$I_D = k(V_{GS} - V_t)^2$$

ALSO:

$$I_S = I_D, \quad I_G = 0 \quad (\text{always})$$

GIVEN CASES

(a)	(b)	(c)	(d)
$V_{GS} = 2 \text{ V}$ $V_{DS} = 10 \text{ V}$	$V_{GS} = 5 \text{ V}$ $V_{DS} = 0 \text{ V}$	$V_{GS} = 5 \text{ V}$ $V_{DS} = 1 \text{ V}$	$V_{GS} = 5 \text{ V}$ $V_{DS} = 6 \text{ V}$

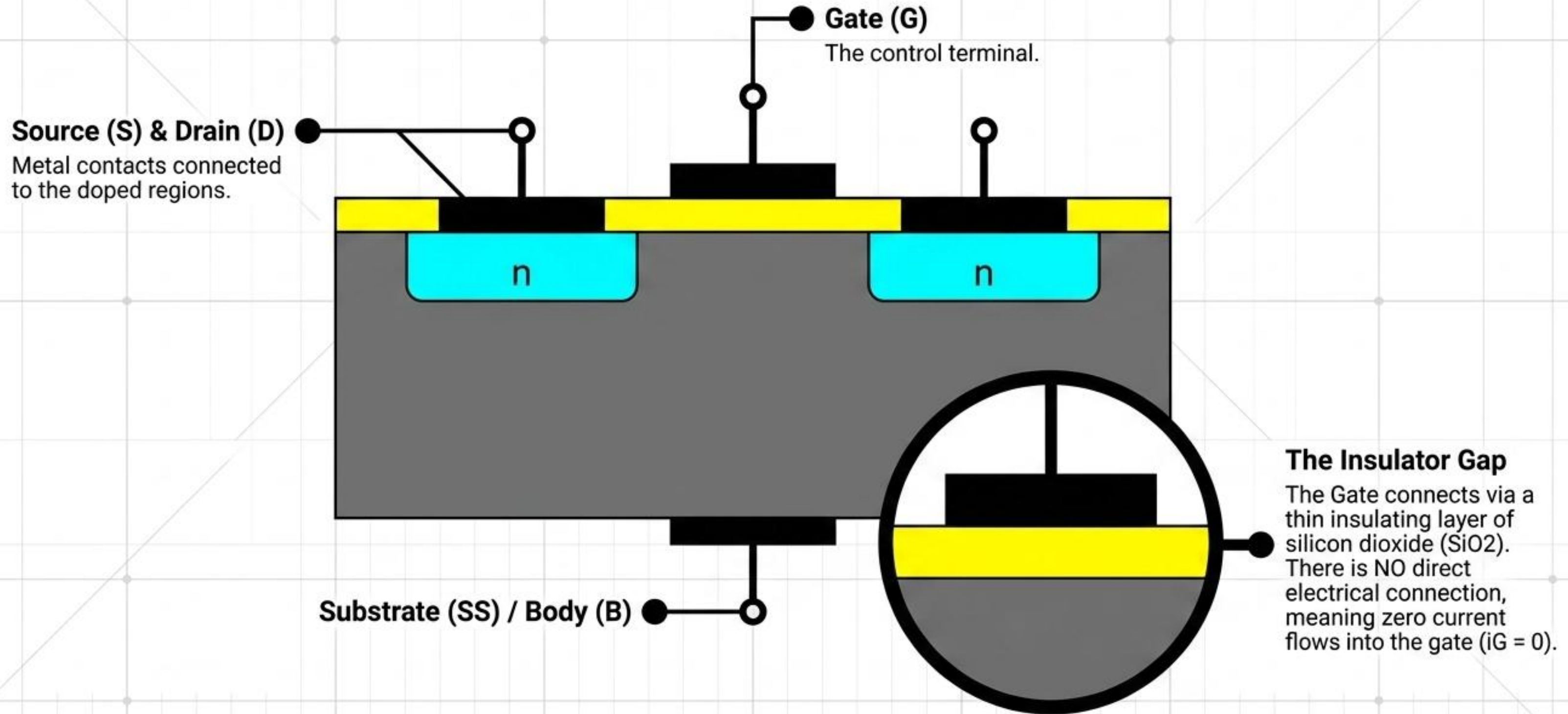
SOLUTION

(a)	(b)	(c)	(d)
$V_{GS} = 2 \text{ V} < V_t = 3 \text{ V}$ $\Rightarrow$ Cutoff Region	$V_{GS} = 5 \text{ V} > V_t = 3 \text{ V}$ BUT $V_{DS} = 0 \text{ V}$ $\Rightarrow$ OFF Region	$V_{GS} = 5 \text{ V} > V_t = 3 \text{ V}$ $V_{DS} = 1 \text{ V} < V_{GS} - V_t = 2 \text{ V}$ $\Rightarrow$ Triode Region	$V_{GS} = 5 \text{ V} > V_t = 3 \text{ V}$ $V_{DS} = 6 \text{ V} \geq V_{GS} - V_t = 2 \text{ V}$ $\Rightarrow$ Saturation Region
$I_D = 0 \text{ mA}$	$I_D = 0 \text{ mA}$	$I_D = 2(1) \left[(2)(1) - \frac{1}{2}(1)^2 \right]$ $= 2(2 - 0.5) = 3 \text{ mA}$	$I_D = k(V_{GS} - V_t)^2$ $= 1(2)^2 = 4 \text{ mA}$
$I_S = 0 \text{ mA}$	$I_S = 0 \text{ mA}$	$I_S = 3 \text{ mA}$	$I_S = 4 \text{ mA}$
$I_G = 0 \text{ A}$	$I_G = 0 \text{ A}$	$I_G = 0 \text{ A}$	$I_G = 0 \text{ A}$

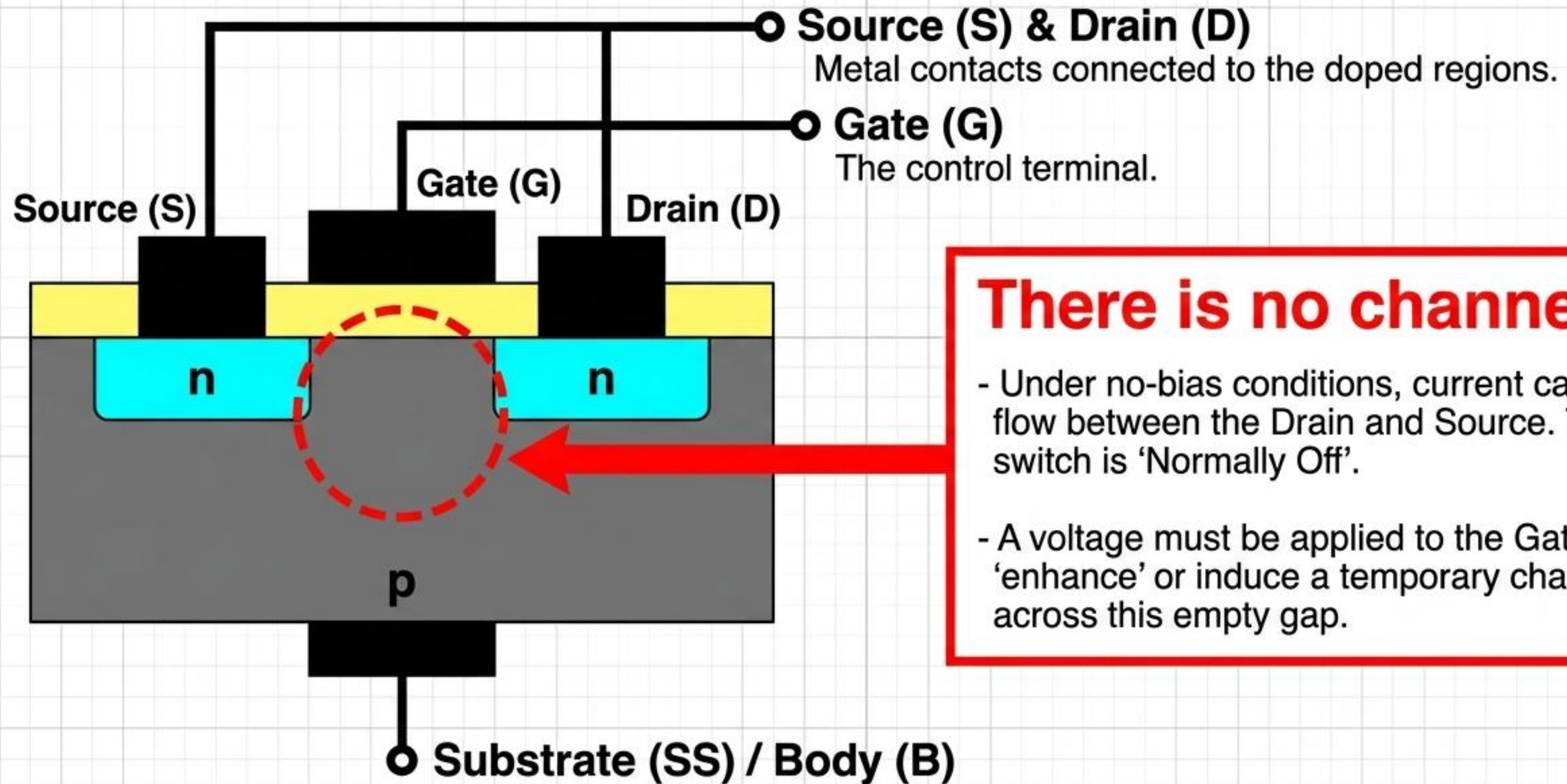
SUMMARY TABLE

Case	(a)	(b)	(c)	(d)
Region	Cutoff	OFF ($V_{DS} = 0$)	Triode	Saturation
I_D	0 mA	0 mA	3 mA	4 mA
I_S	0 mA	0 mA	3 mA	4 mA
I_G	0 A	0 A	0 A	0 A

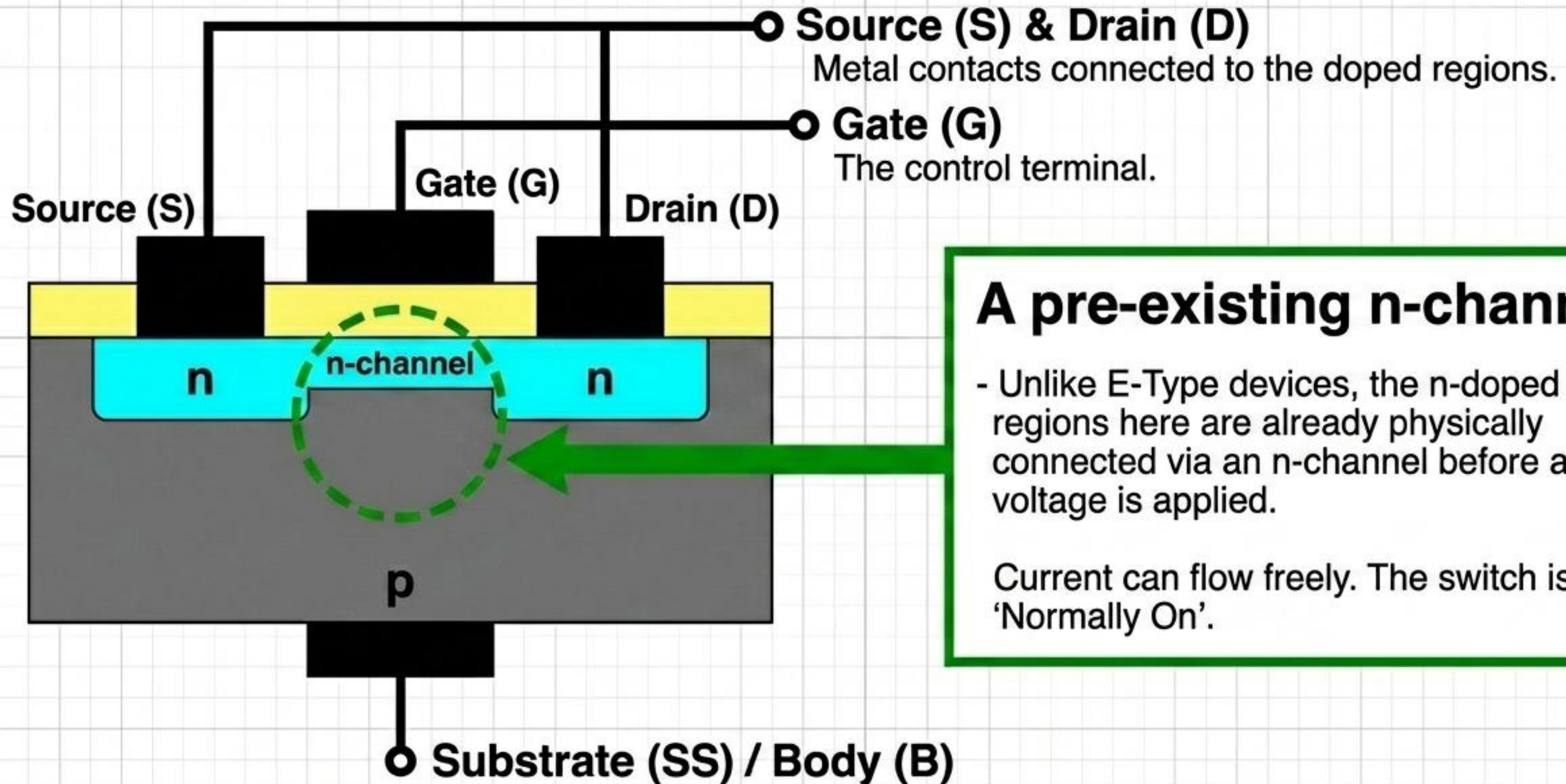
The Anatomy of a Transistor



Enhancement-Type (E-Type): The 'Normally Off' Switch



Depletion-Type (D-Type): The “Normally On” Switch



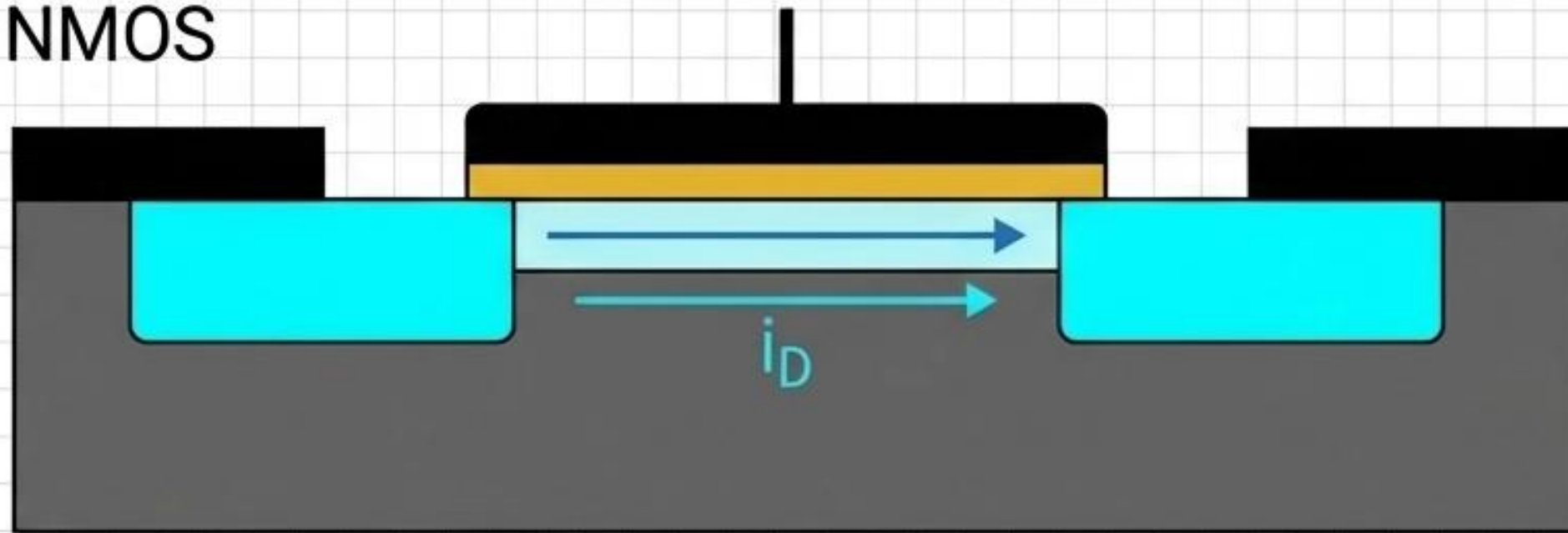
A pre-existing n-channel.

- Unlike E-Type devices, the n-doped regions here are already physically connected via an n-channel before any voltage is applied.

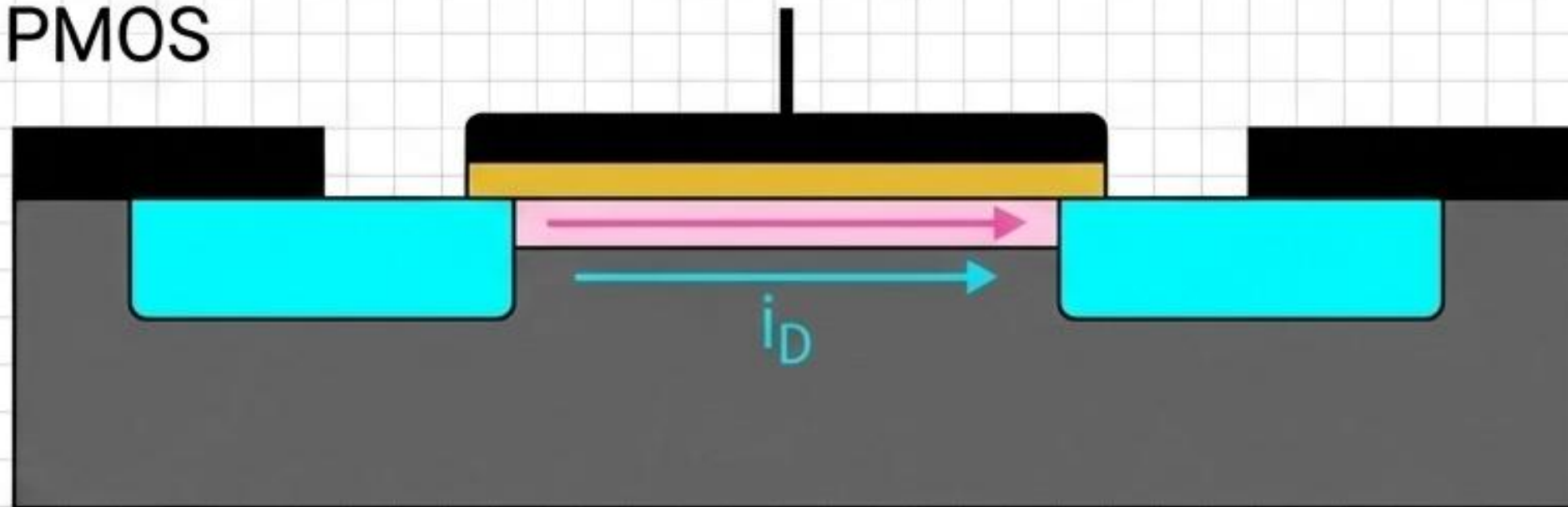
Current can flow freely. The switch is 'Normally On'.

Material Polarity: NMOS vs. PMOS

NMOS



PMOS

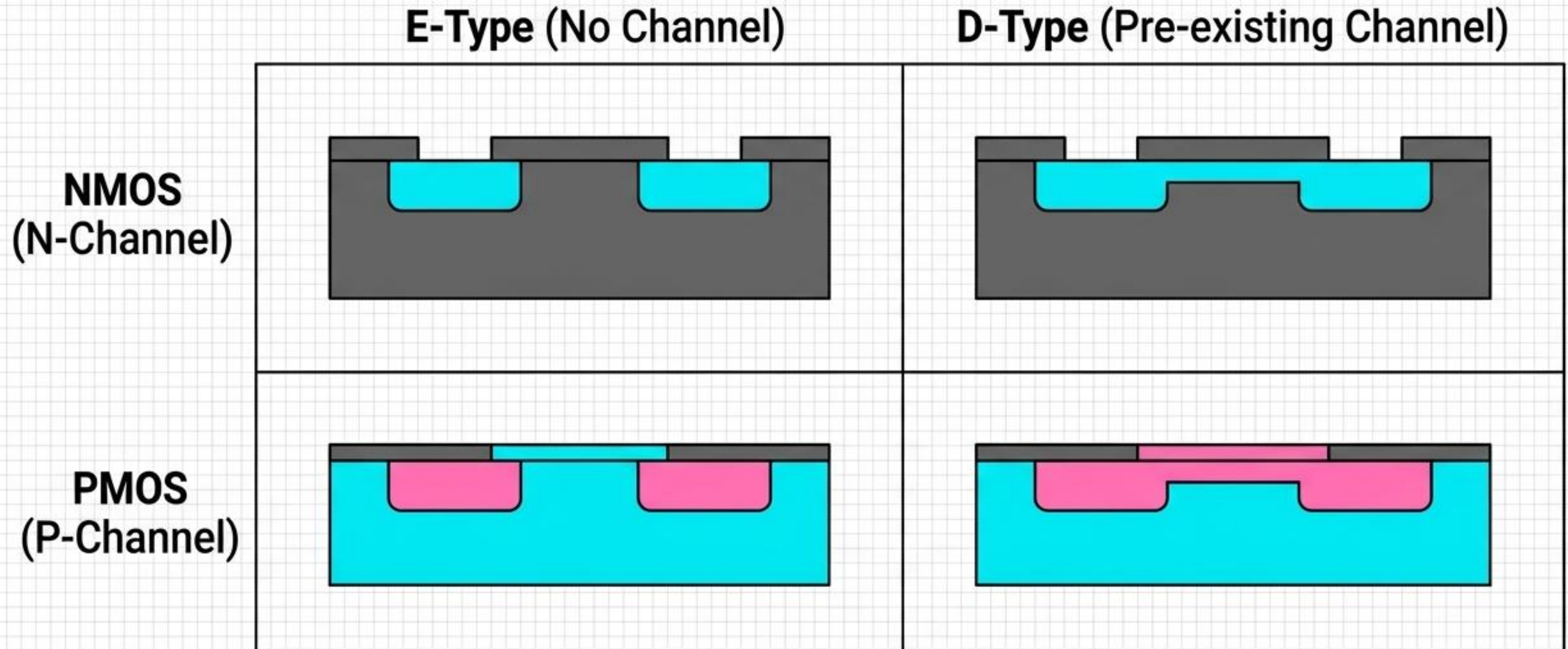


Total Polarity Reversal

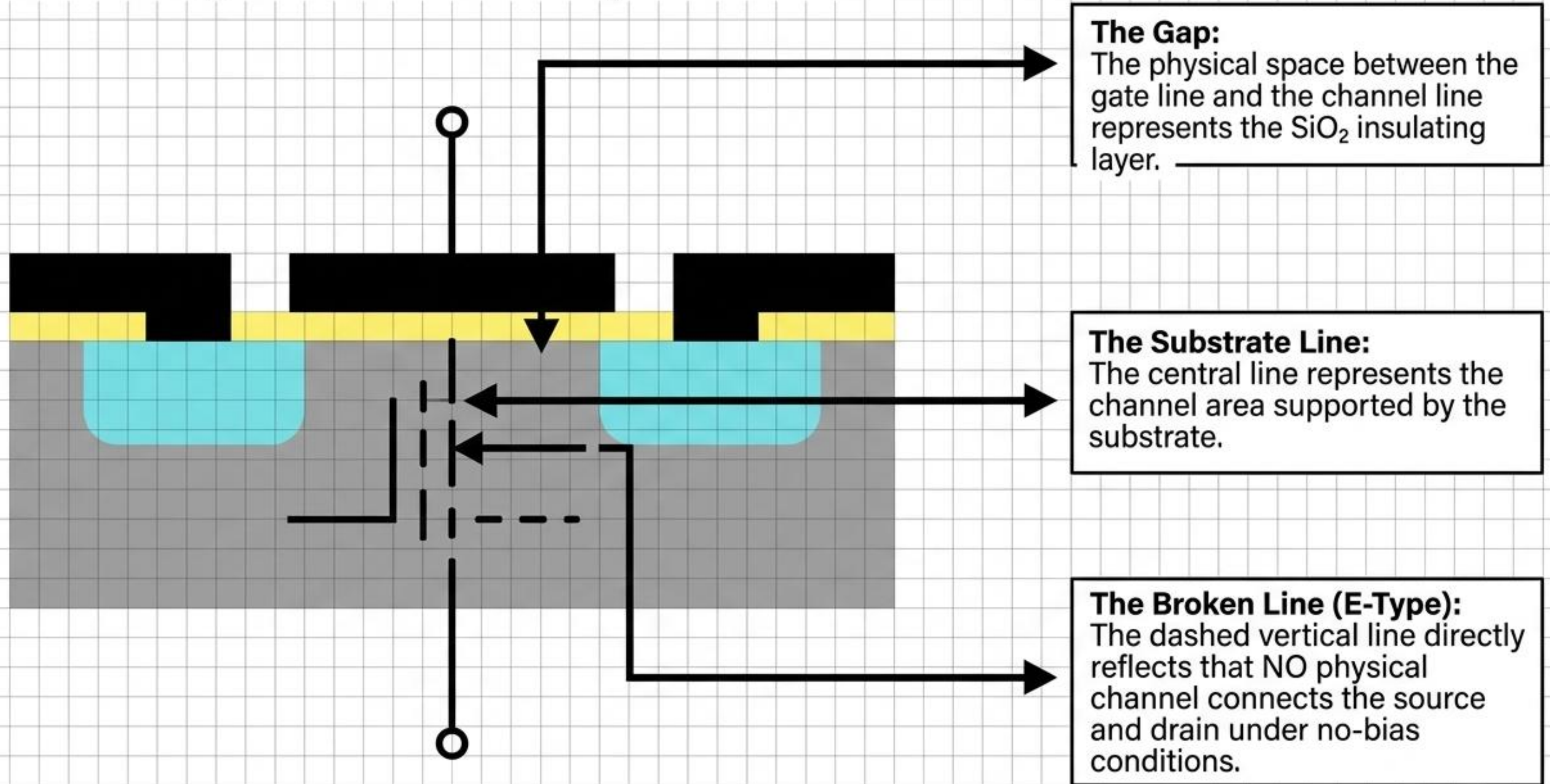
PMOS transistors are structurally identical to NMOS transistors—except that all semiconductor materials are exactly reversed in polarity. N-type material becomes P-type, and vice versa.

The Four MOSFET Families

A diagnostic matrix combining structural duality (channel presence) and material duality (polarity).

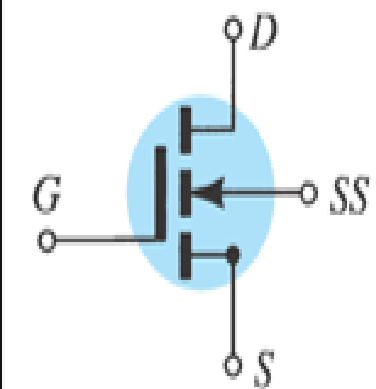


Translating Silicon into Symbols

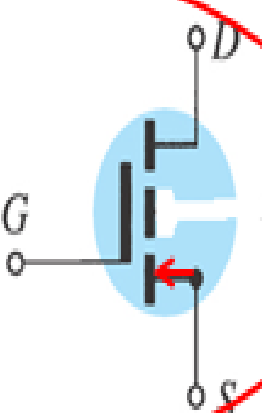
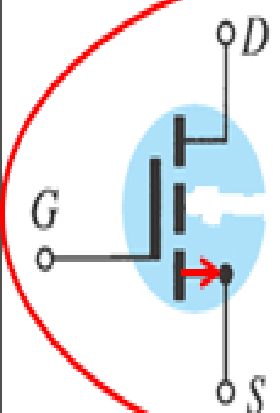
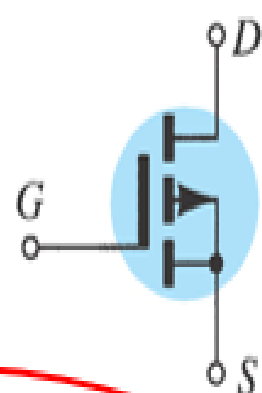
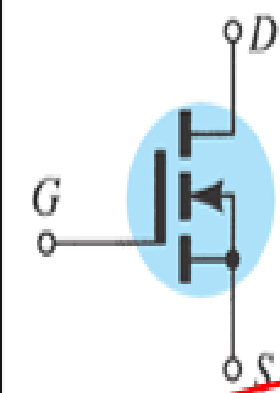
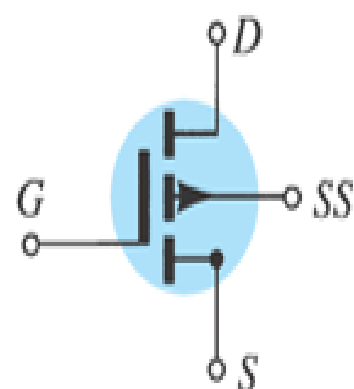


MOSFET Symbols

n-channel



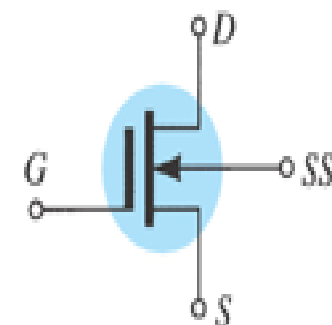
p-channel



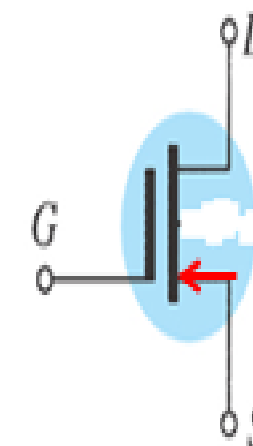
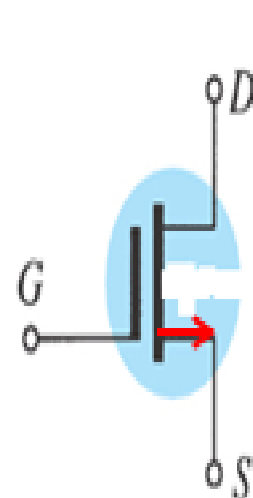
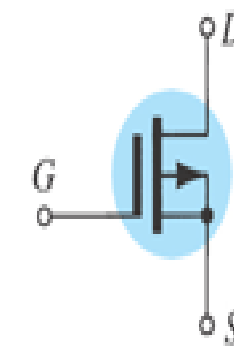
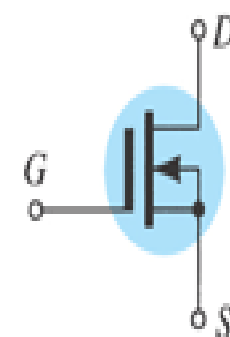
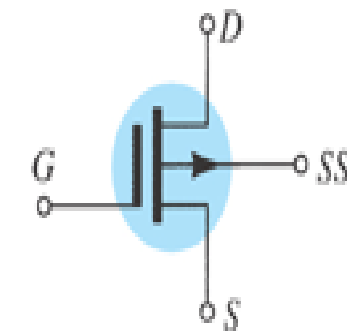
- The symbols try to reflect the actual construction of the device.
- The dashed line between drain and source is chosen to reflect the fact that a channel does not exist between the two under no-bias conditions.
- It is, in fact, the only difference between the symbols for the depletion-type and enhancement-type MOSFETs.
- NMOS: source current i_S leaves the NMOS.
- PMOS: source current i_S enters the PMOS.

D-Type MOSFET Symbols

n-channel



p-channel



- The symbols try to reflect the actual construction of the device.
- The lack of a direct connection (due to the gate insulation) between the gate and the channel is represented by a space between the gate and the other terminals of the symbol.
- The vertical line representing the channel is connected between the drain and the source and is "supported" by the substrate.
- Two symbols are provided for each type of channel to reflect the fact that in some cases the substrate is externally available, whereas in others it is not.
- For most of our analysis, the substrate and the source will be connected and the lower symbols will be employed.

The CMOS Revolution

CMOS (Complementary MOS) is the most widely used IC technology, pairing a p-channel (PMOS) and n-channel (NMOS) device on a single substrate to work cooperatively.

Why CMOS replaced BJT in digital logic:



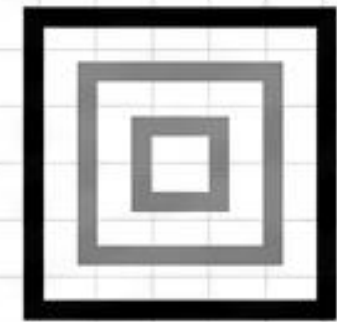
1. Less Power

Uses near-zero static power. Current only flows during state switching, drastically reducing heat generation.



2. Higher Impedance

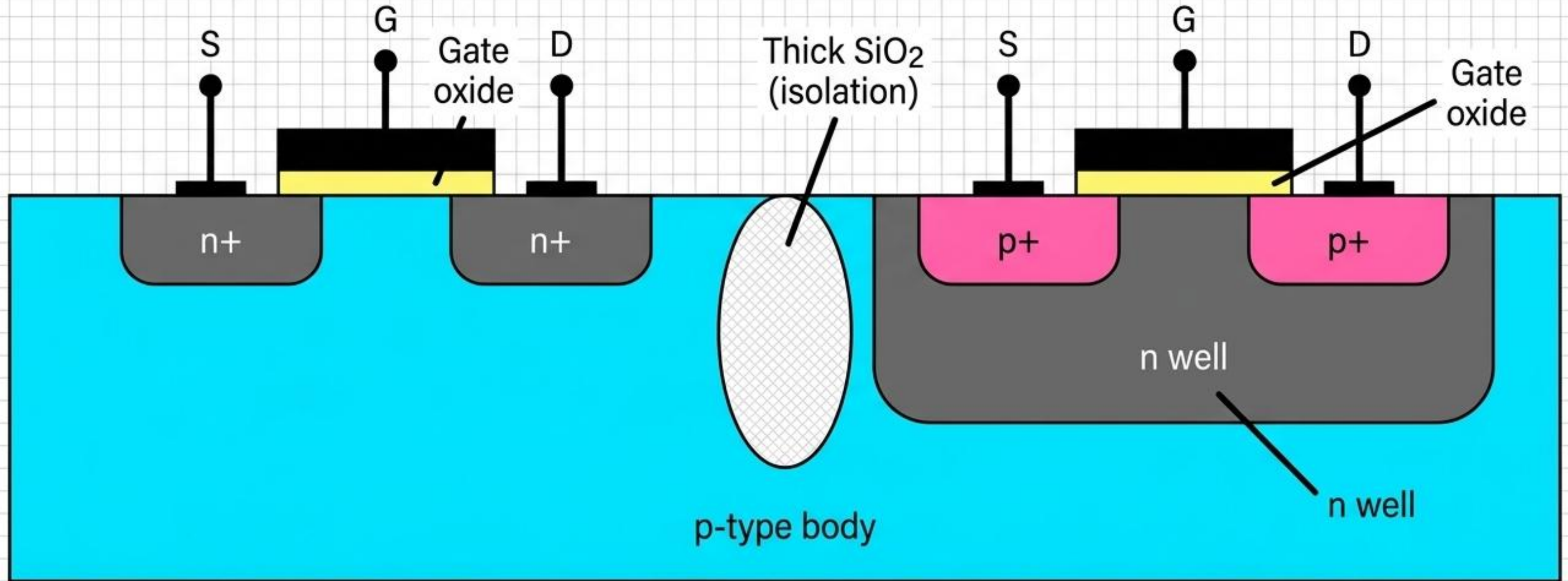
Excellent electrical isolation between the gate and the rest of the circuit.



3. Easier Scaling

Can be miniaturized and densely packed more effectively onto modern silicon microchips.

Physical CMOS Architecture

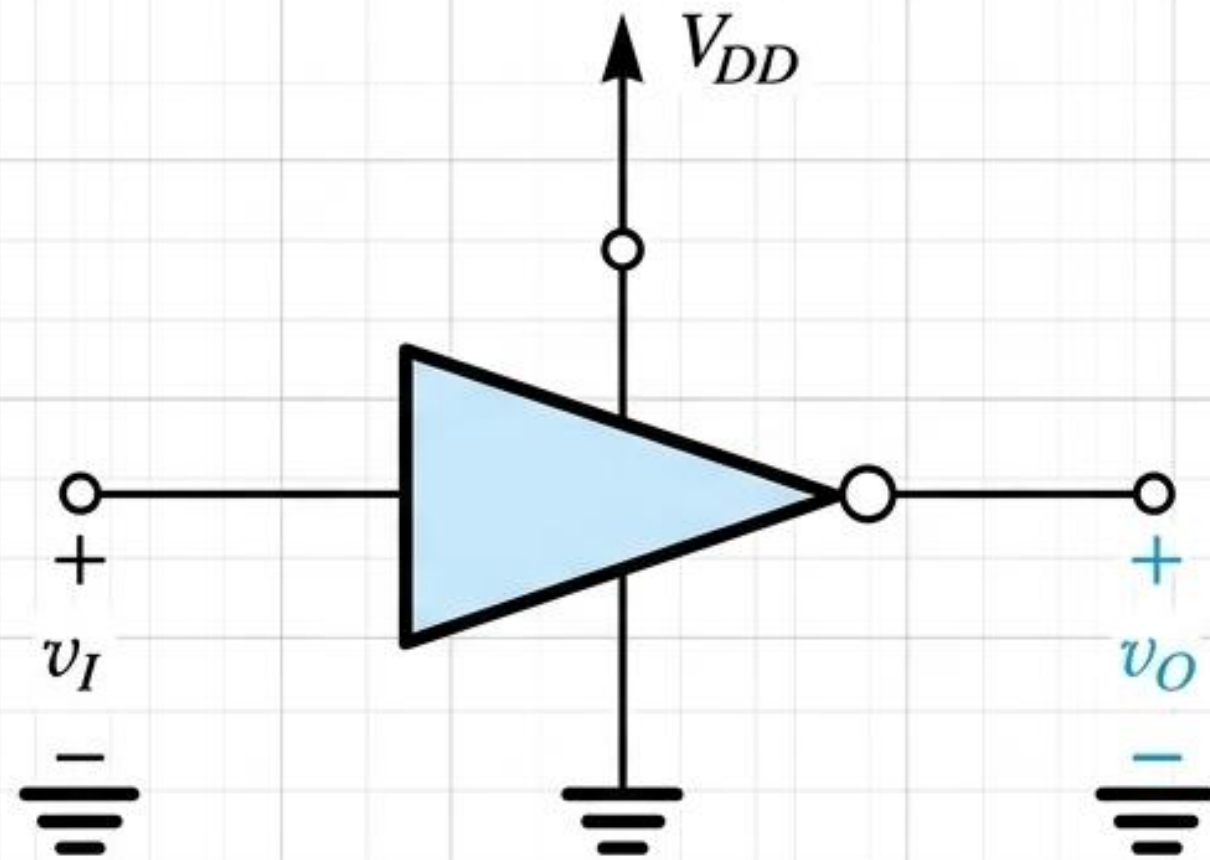


Both devices are fabricated onto the exact same substrate. An 'n-well' must be created as a localized environment to house the PMOS transistor within the overall p-type body.

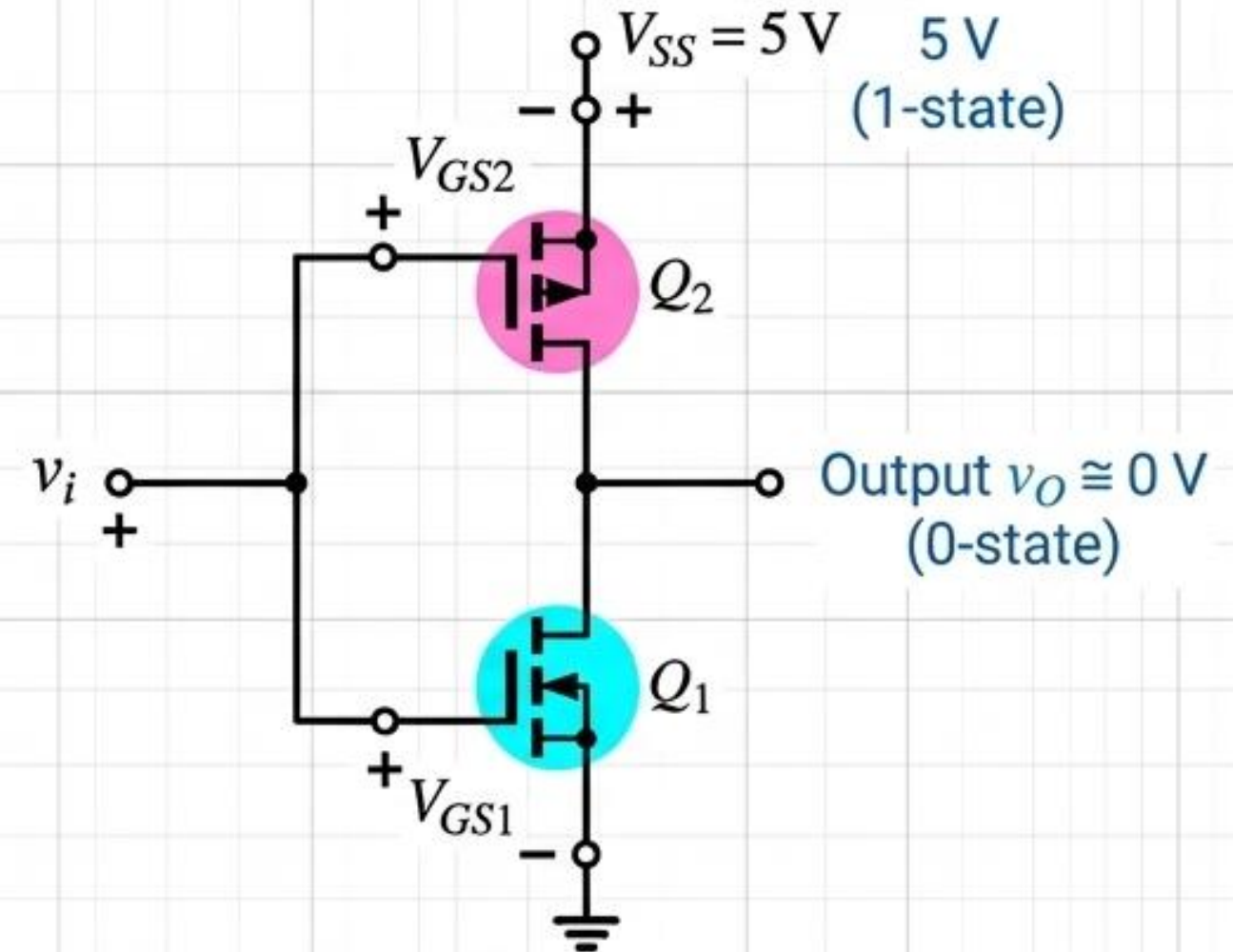
The Logic Inverter: Building Digital 0s and 1s

The Inverter is the most basic, fundamental element in digital circuit design.

Abstract Logic Symbol



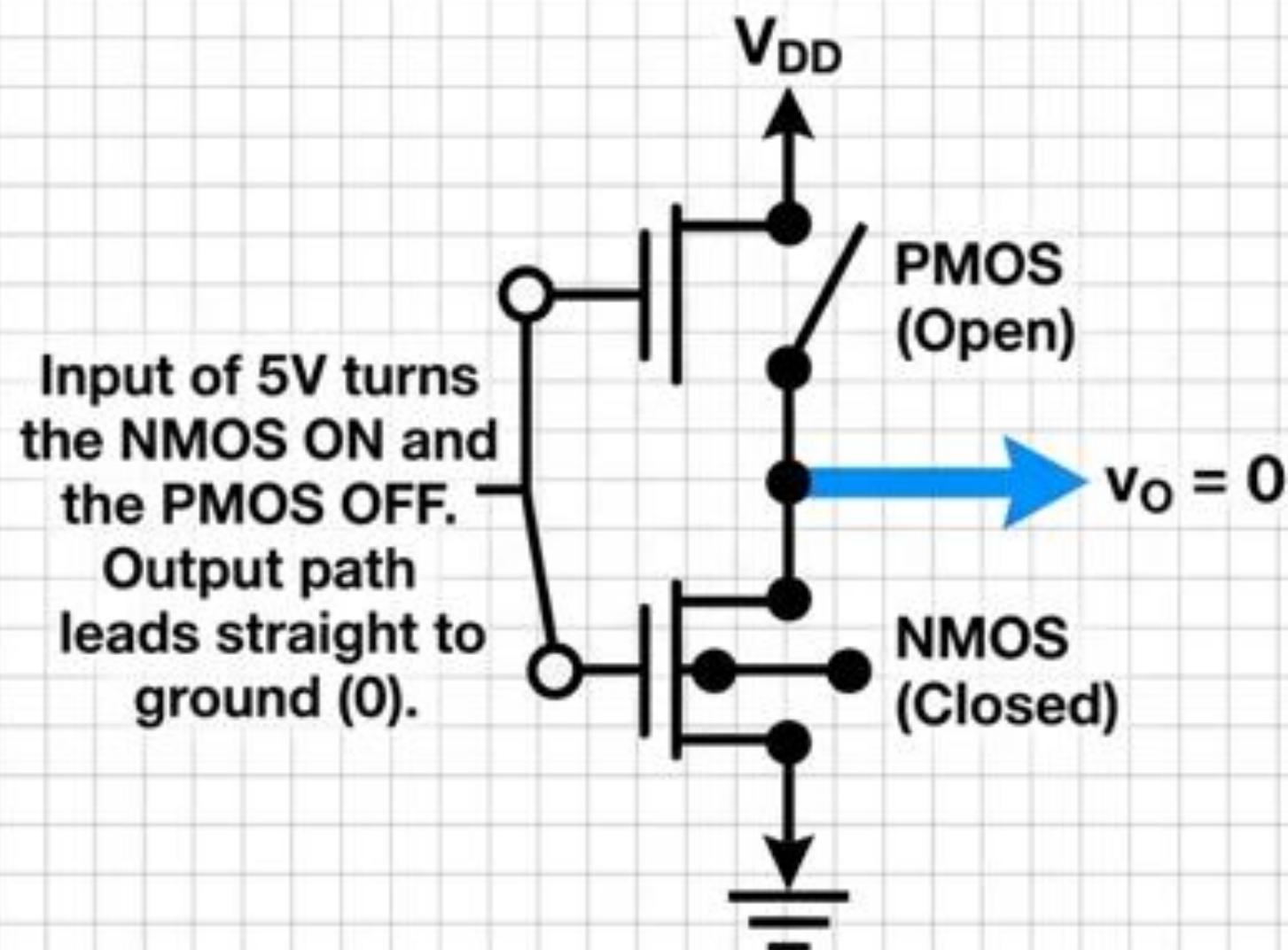
CMOS Circuit Implementation



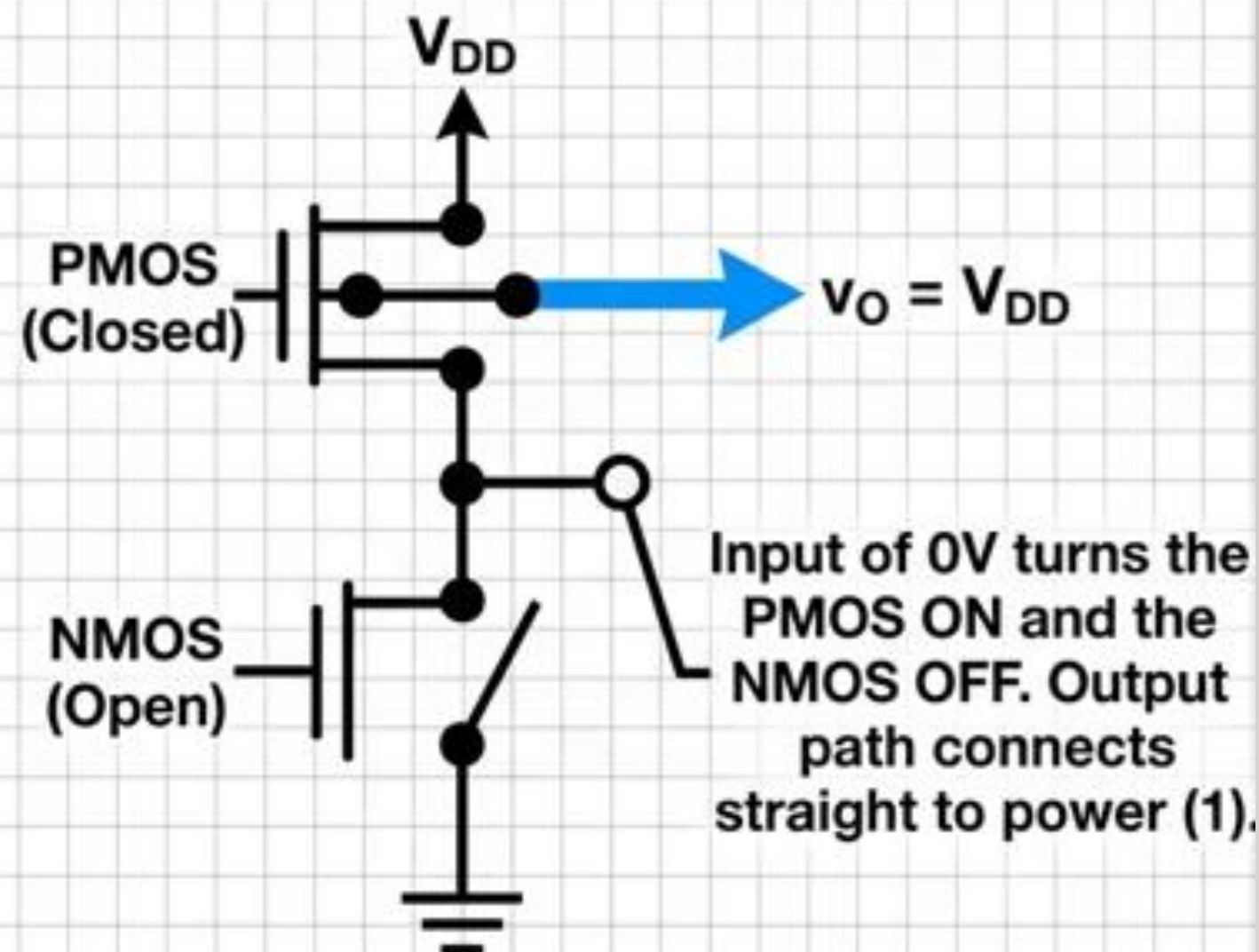
The Engineering Goal: An Input of 1 (5V) forces an Output of 0 (0V), and an Input of 0 forces an Output of 1. By combining NMOS and PMOS, we create a foolproof binary switch.

How CMOS Logic Works

Input 1 -> Output 0



Input 0 -> Output 1

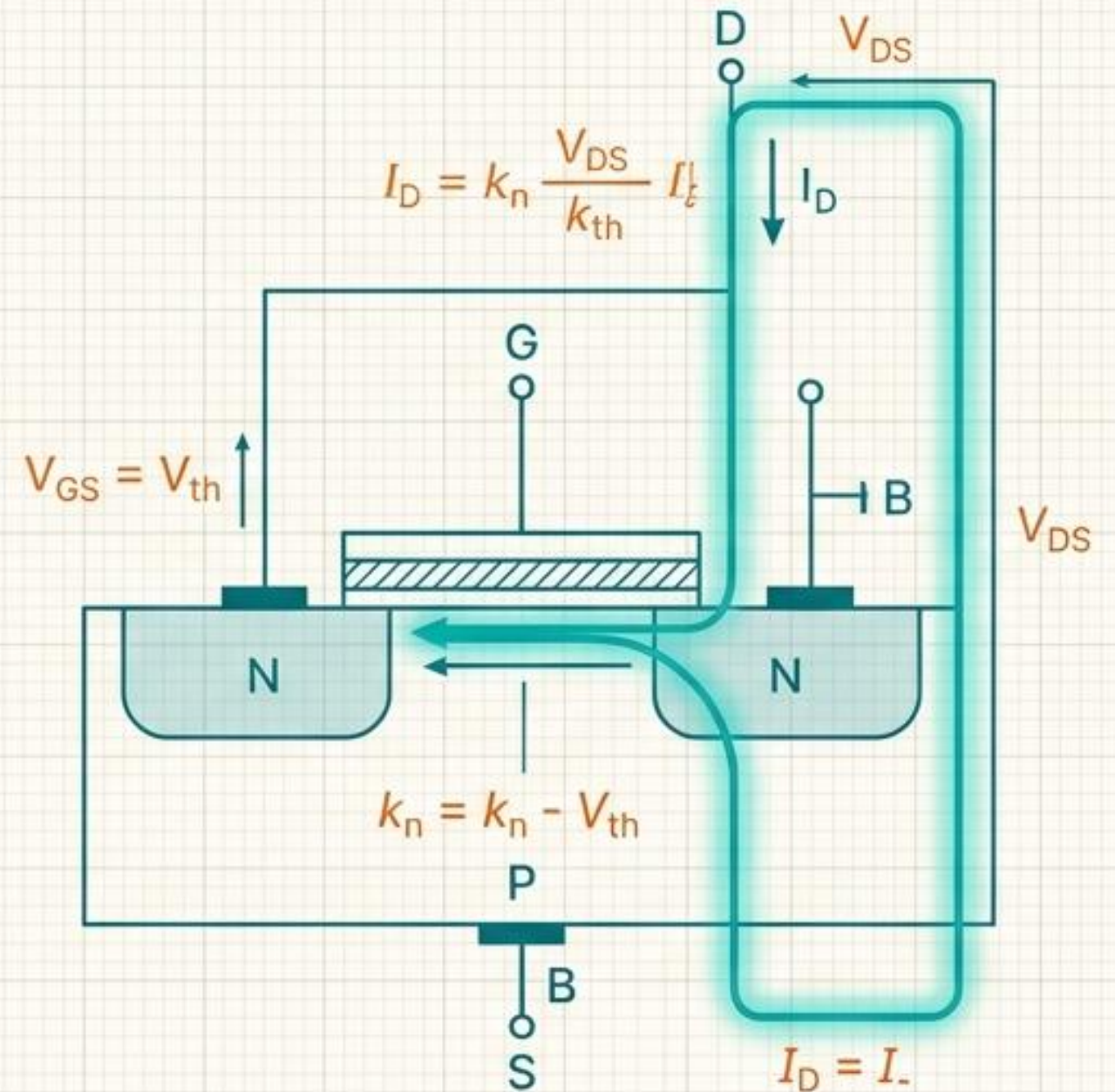


Because NMOS and PMOS react to opposite voltages, one switch is always open while the other is closed. This perfect “complementary” teamwork prevents short circuits and makes modern digital logic possible with nearly zero wasted power.

E-MOSFET DC Biasing: The Graphical Blueprint

A visual framework for solving non-linear circuit analysis.

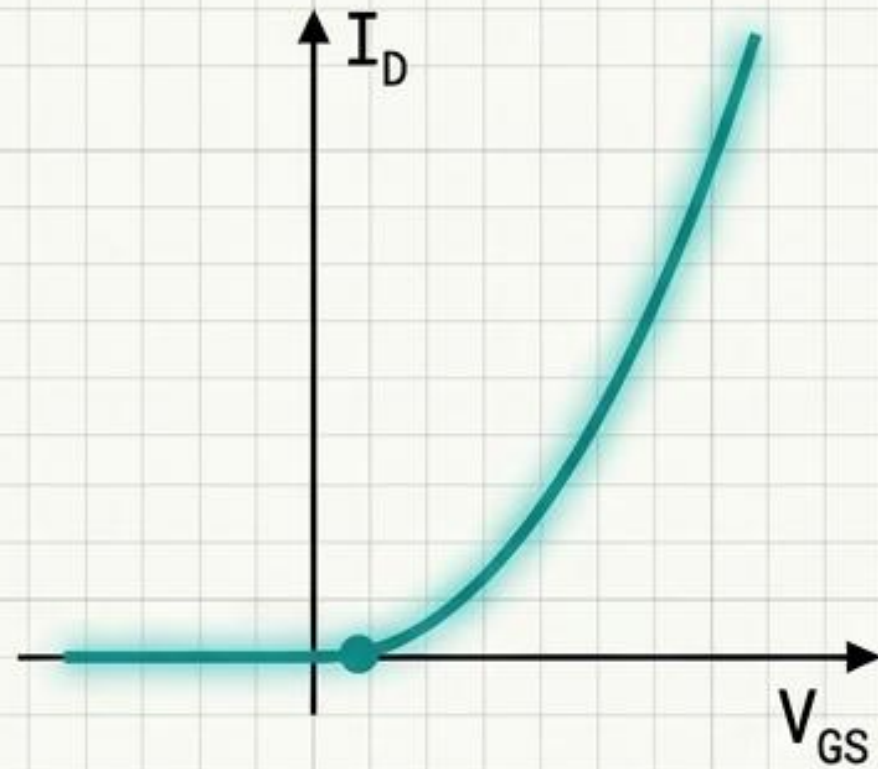
Bypassing complex algebra with physical modeling.



	The Mathematical Approach	The Graphical Approach
The Core Challenge	Non-linear relationship between input V_{GS} and output I_D	Plots intersecting constraints physically
Calculation Speed	Slow (requires solving complex quadratic equations)	Quick (requires only 3-4 plotting points)
Precision Level	Highly accurate (algebraically exact)	Practically accurate (subject to visual estimation)
Engineering Application	Theoretical precision modeling	Rapid prototyping and amplifier design

We design amplifiers using the graphical solution: trading perfect algebraic accuracy for immediate, practical visualization.

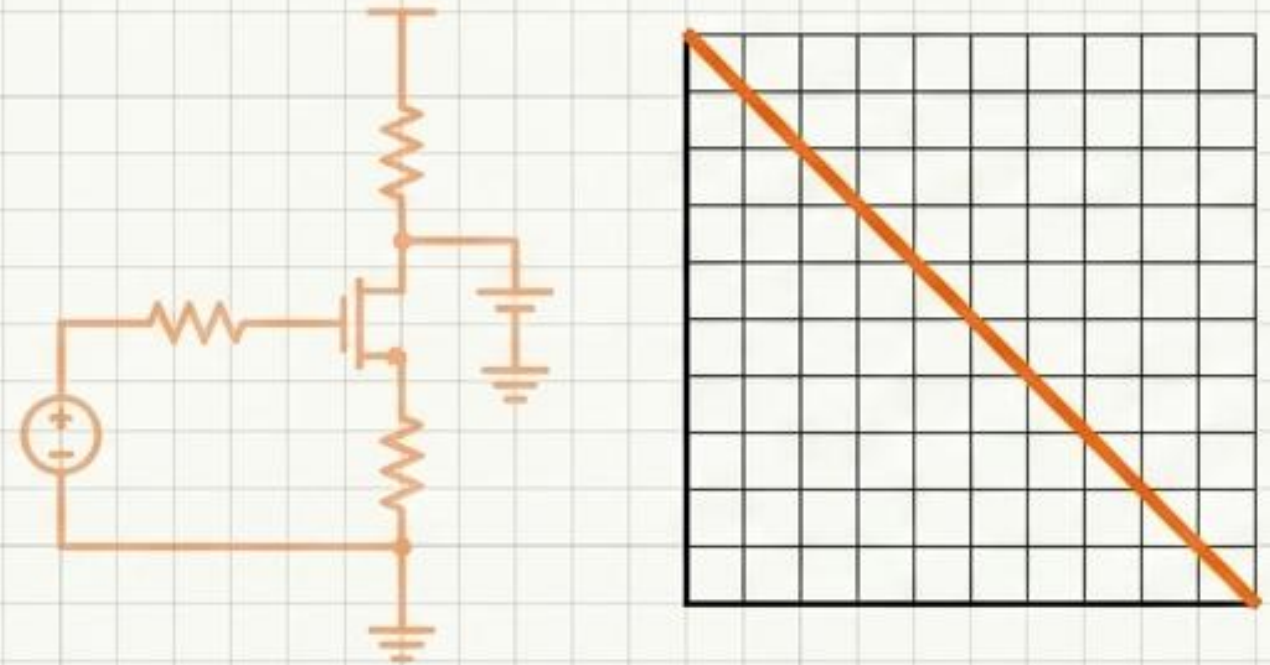
THE DEVICE (Physics)



$$\begin{aligned}I_G &\cong 0 \text{ A} \\I_D &= I_S \\I_D &= k(V_{GS} - V_{Th})^2\end{aligned}$$

The absolute physical constraints of the E-MOSFET. It *must* operate on this non-linear curve.

THE NETWORK (Environment)



Kirchhoff's Voltage Law (KVL)
Ohm's Law

The external constraints forced upon the device by resistors and power supplies. It *must* operate on this linear path.

PLOTTING THE TRANSFER CURVE: A 3-STEP SYNTHESIS

DATASHEET SPEC

$$V_{GS(Th)} = [\text{Value}] \text{ V}$$

$$I_{D(on)} = [\text{Value}] \text{ mA}$$

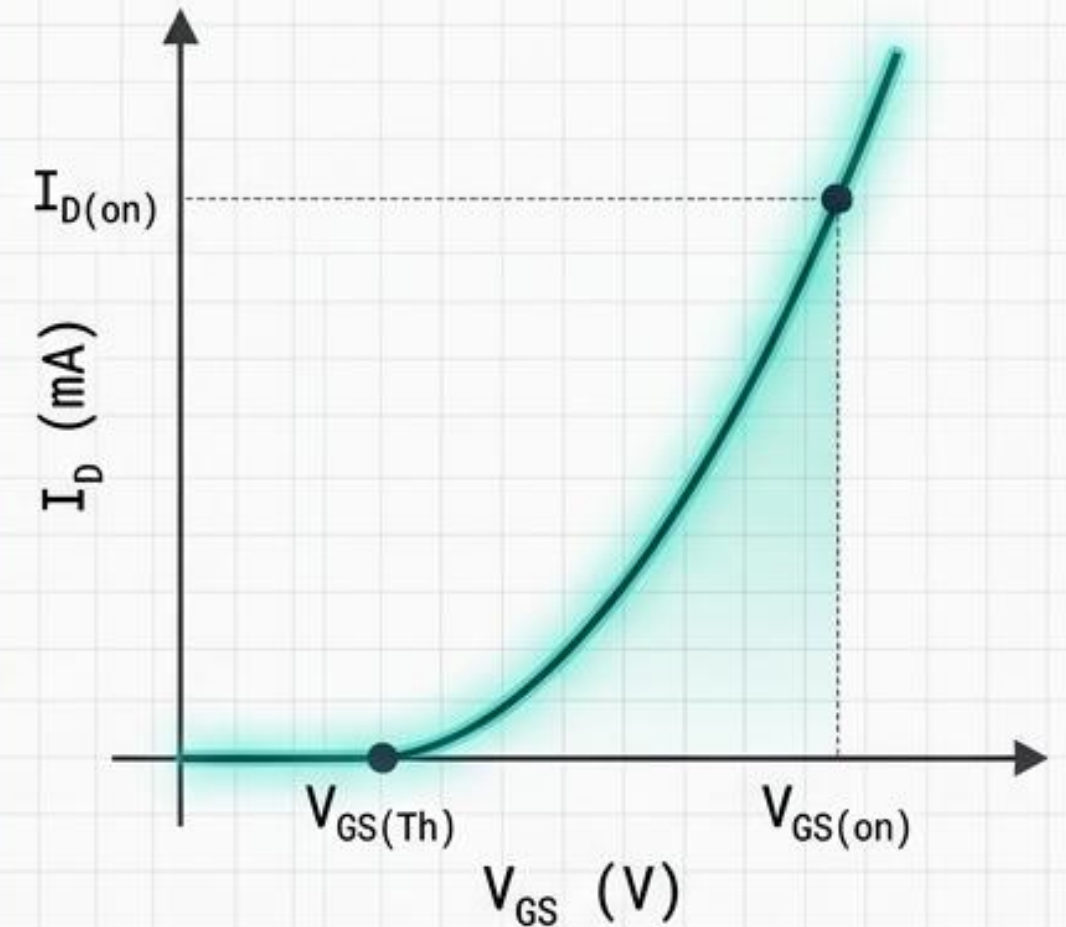
at $V_{GS(on)} = [\text{Value}] \text{ V}$

STEP 2: CALCULATE THE DEVICE CONSTANT (k)

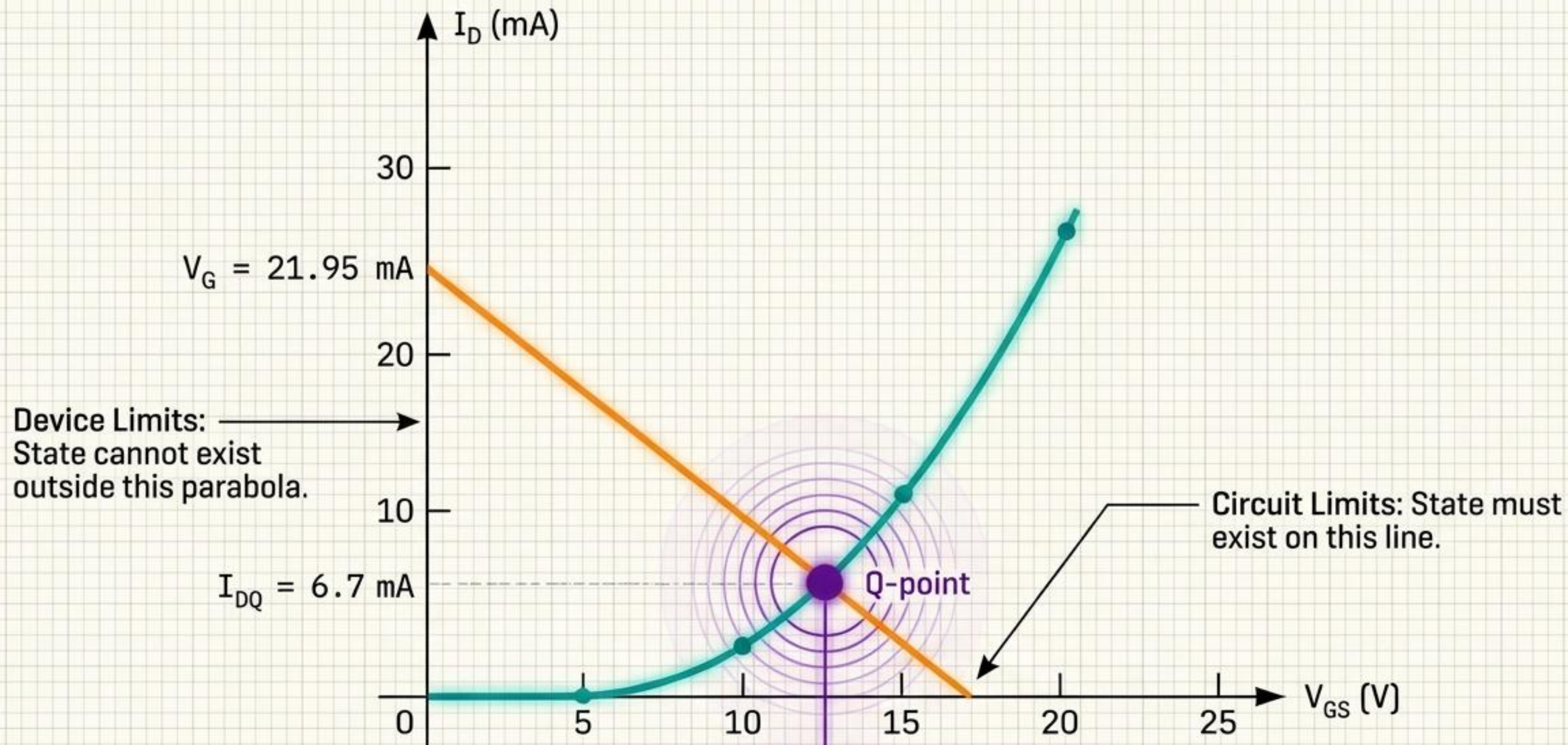
$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{GS(Th)})^2}$$

Derive k from known
datasheet points.

STEP 3: GENERATE & PLOT THE CURVE

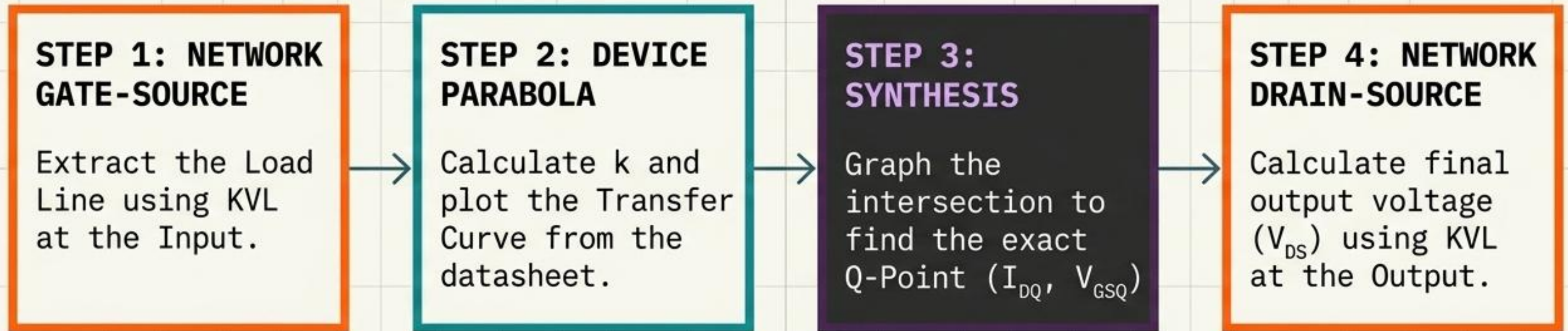


The k constant acts as the device's DNA. It scales the universal physics of a field-effect transistor to the specific manufacturing geometry of the exact component you are using.

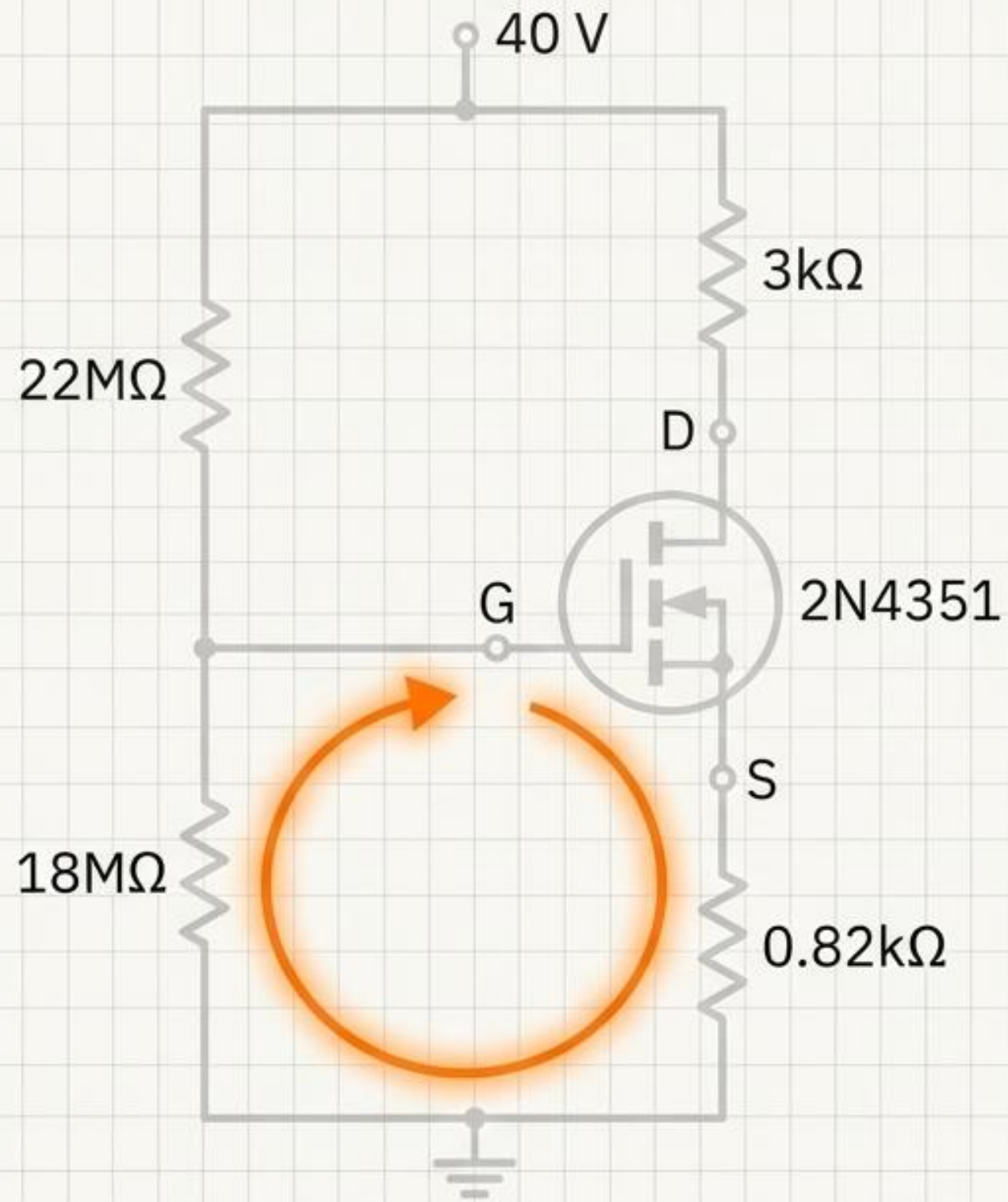


The Q-Point (Quiescent Point).
The single mathematical state where the laws of device physics and the laws of circuit network topology agree.

THE E-MOSFET BIASING STANDARD OPERATING PROCEDURE



This 4-step physical process scales to *any* E-MOSFET biasing configuration, no matter how complex the surrounding network.



Case Study 1: Step 1 (Network KVL)

$$V_{DD} = 40V$$

$$R_1 = 22M\Omega$$

$$R_2 = 18M\Omega$$

$$R_S = 0.82k\Omega.$$

Calculate base voltage $V_G = 18V$.

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2} = \frac{(18 M\Omega)(40 V)}{22 M\Omega + 18 M\Omega} = 18 V$$

Final load line:

$$V_{GS} = V_G - I_D(R_S) \rightarrow V_{GS} = 18V - I_D(0.82k\Omega)$$

Case Study 1: Step 2 (Device Parabola)

Component: 2N4351

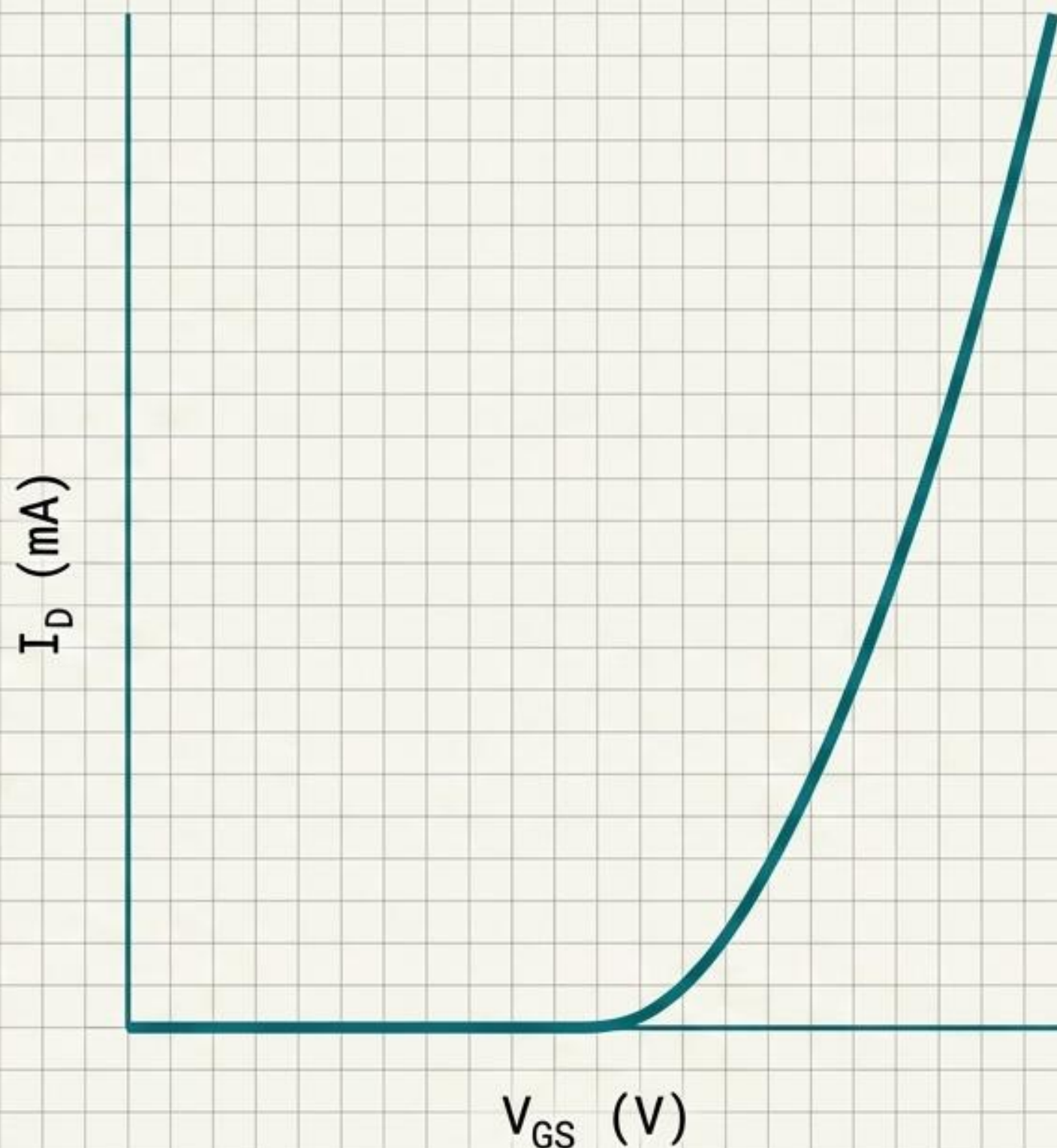
$$V_{GS(Th)} = 5V$$

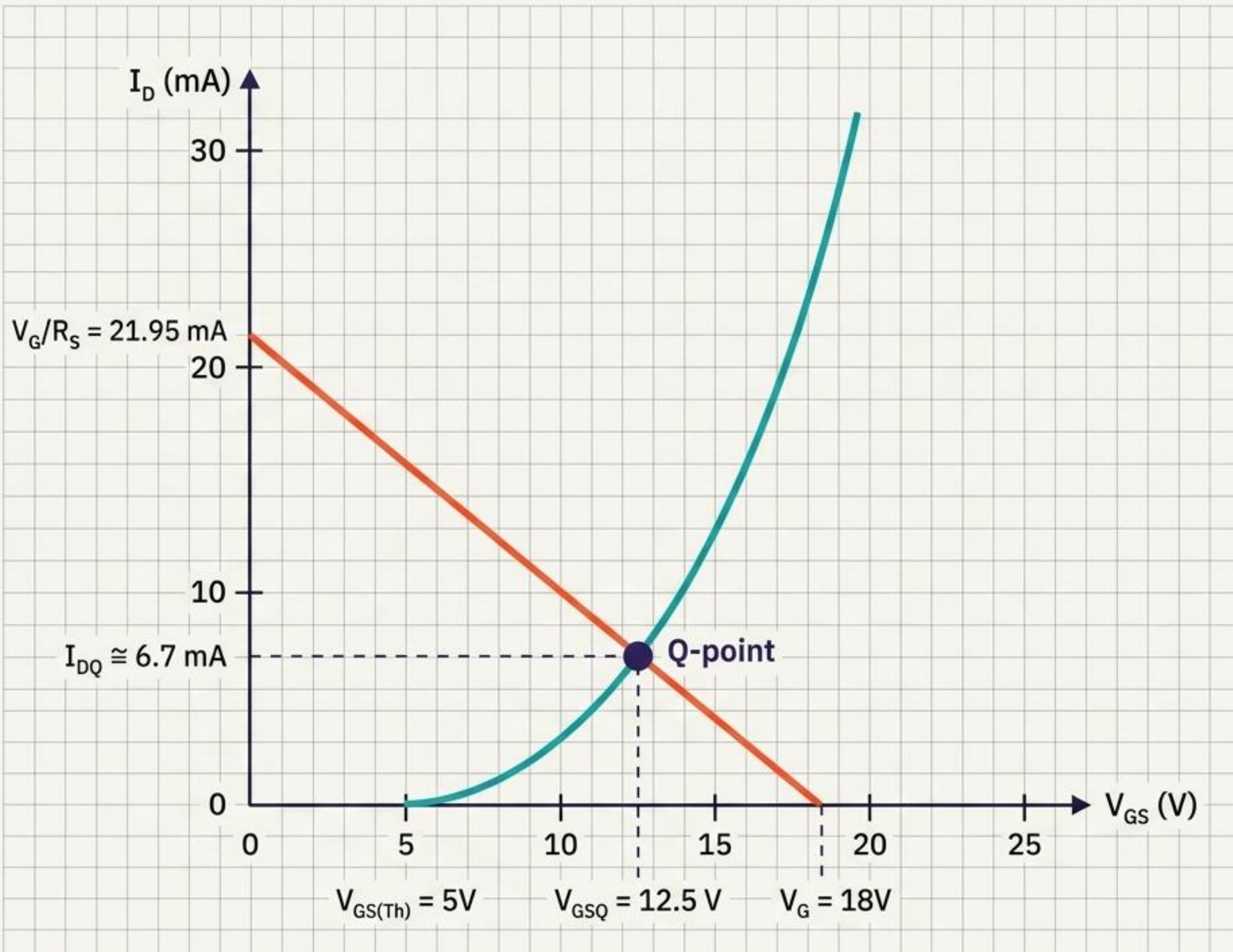
$$I_{D(on)} = 3mA \text{ at } V_{GS(on)} = 10V$$

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{GS(Th)})^2}$$

$$k = \frac{3 \text{ mA}}{(10 \text{ V} - 5 \text{ V})^2} = 0.12 \times 10^{-3} \text{ A/V}^2$$

$$I_D = 0.12 \times 10^{-3} (V_{GS} - 5)^2$$





Case Study 1: Step 3 (Synthesis)

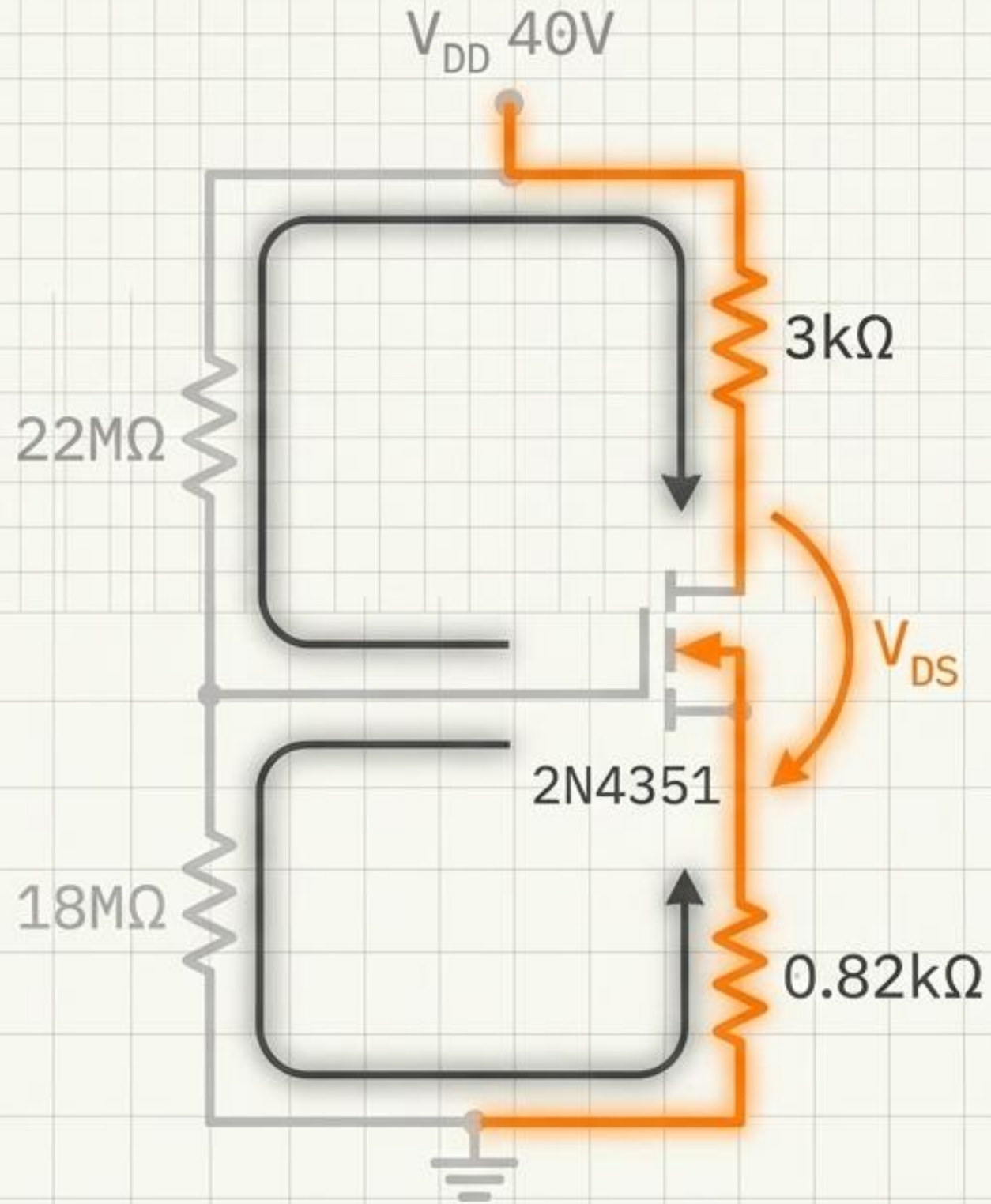
Reading the graph reveals the quiescent operating constraints.

1

$$I_{DQ} \cong 6.7 \text{ mA}$$

2

$$V_{GSQ} \cong 12.5 \text{ V}$$



Case Study 1: Step 4 (Output KVL)

$$V_{DS} = V_{DD} - I_D(R_S + R_D)$$

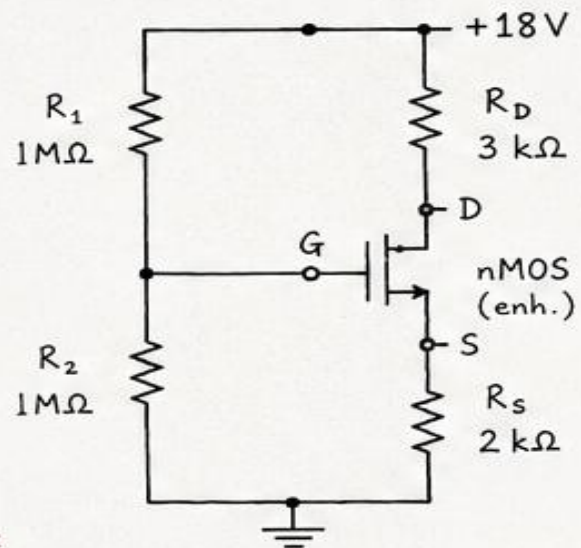
Substitute the resolved $I_{DQ}(6.7\text{mA})$ into the equation.

Final Output:

$$V_{DS} = 14.4\text{V}$$

EXAM QUESTION: E-MOSFET DC BIASING (Graphical Method)

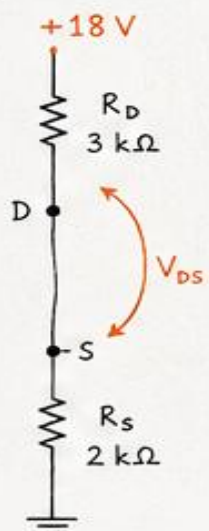
An n-channel enhancement MOSFET is connected in the voltage-divider bias circuit below.



Required:

- 1) Determine V_G .
- 2) Derive the network load-line equation relating I_D and V_{GS} .
- 3) Find the two intercepts to draw the load line.
- 4) Calculate the MOSFET constant k .
- 5) Generate points for the MOSFET transfer characteristic.
- 6) Plot the load line and device curve and determine I_{DQ} and V_{GSQ} .
- 7) Calculate V_{DSQ} .
- 8) Verify that the MOSFET operates in saturation.

7) Calculate V_{DSQ} .



KVL around the drain-source loop:

$$V_{DD} = I_D R_D + V_{DS} + I_D R_S$$

$$V_{DS} = V_{DD} - I_D (R_D + R_S)$$

Substitute $I_{DQ} = 2 \text{ mA}$:

$$V_{DSQ} = 18 - (2 \text{ mA})(3 \text{ k}\Omega + 2 \text{ k}\Omega)$$

$$= 18 - (2 \text{ mA})(5 \text{ k}\Omega)$$

$$= 18 - 10$$

$$V_{DSQ} = 8 \text{ V}$$

8) Verify saturation.

For saturation:

$$V_{DS} \geq V_{GS} - V_{th}$$

At Q-point:

$$V_{GSQ} - V_{th} = 5 - 3 = 2 \text{ V}$$

$$V_{DSQ} = 8 \text{ V}$$

$$\text{Since } 8 \text{ V} > 2 \text{ V}$$

MOSFET is in saturation. ✓

Given:

$$V_{DD} = 18 \text{ V}$$

$$R_1 = 1 \text{ M}\Omega$$

$$R_2 = 1 \text{ M}\Omega$$

$$R_D = 3 \text{ k}\Omega$$

$$R_S = 2 \text{ k}\Omega$$

Datasheet:

$$V_{GS(th)} = 3 \text{ V}$$

$$I_{D(on)} = 4.5 \text{ mA}$$

$$\text{at } V_{GS(on)} = 6 \text{ V}$$

1) Determine V_G .

Since $I_G \approx 0$, the gate voltage is set by the voltage divider.

$$V_G = V_{DD} \frac{R_2}{R_1 + R_2}$$

$$= 18 \times \frac{(1\text{M})}{1\text{M} + 1\text{M}}$$

$$= 18 \times \frac{1}{2} = 9 \text{ V}$$

2) Derive network load-line equation.

$$V_{GS} = V_G - V_S = V_G - I_D R_S$$

$$V_{GS} = 9 - I_D (2 \text{ k}\Omega)$$

$$V_{GS} = 9 - 2 I_D \quad (I_D \text{ in mA})$$

Equivalent form:

$$I_D = \frac{9 - V_{GS}}{2} \quad (I_D \text{ in mA})$$

3) Find load-line intercepts.

$$(i) I_D = 0 \rightarrow V_{GS} = 9 - 2(0) = 9 \text{ V}$$

$$\text{Point A: } (V_{GS}, I_D) = (9 \text{ V}, 0 \text{ mA})$$

$$(ii) V_{GS} = 0 \rightarrow I_D = \frac{9 - 0}{2} = 4.5 \text{ mA}$$

$$\text{Point B: } (V_{GS}, I_D) = (0 \text{ V}, 4.5 \text{ mA})$$

Draw a straight line between A and B.

This is the **network load line**.

4) Calculate MOSFET constant k .

Using the datasheet point:

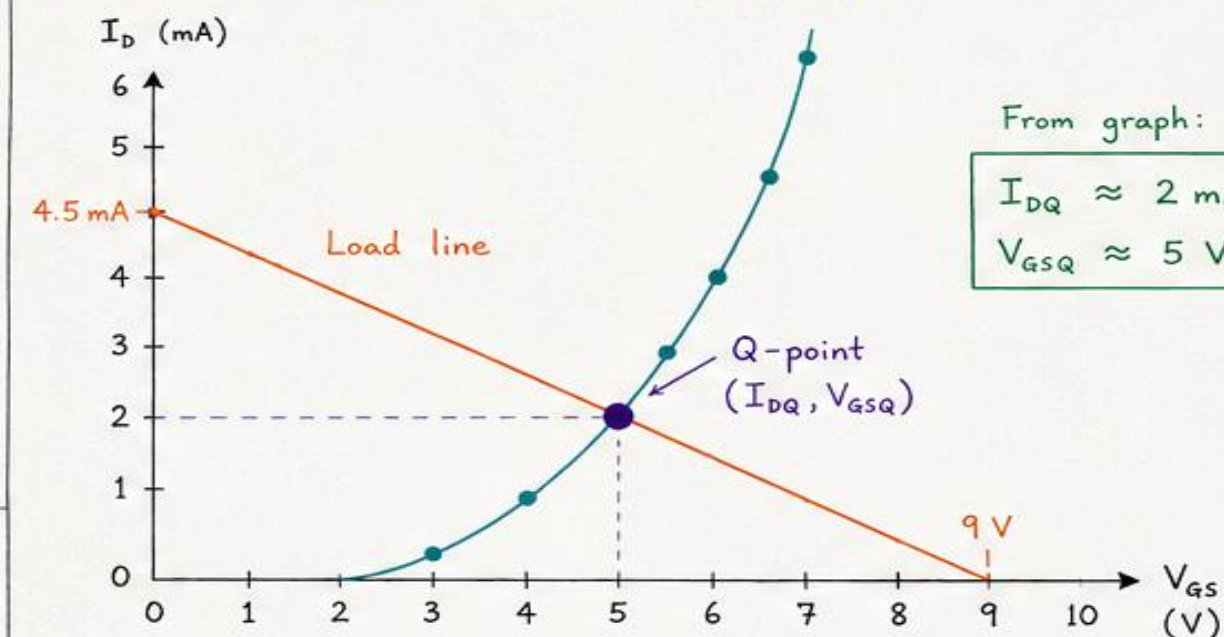
$$I_D = k (V_{GS} - V_{th})^2$$

At $I_{D(on)} = 4.5 \text{ mA}$ when $V_{GS(on)} = 6 \text{ V}$
and $V_{th} = 3 \text{ V}$

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{th})^2}$$
$$= \frac{4.5 \text{ mA}}{(6 - 3)^2} = \frac{4.5}{9}$$

$$k = 0.5 \text{ mA/V}^2$$

6) Plot load line & device curve. Find Q-point.



5) Generate points for device curve.

$$I_D = 0.5 (V_{GS} - 3)^2 \quad (\text{mA})$$

$V_{GS} \text{ (V)}$	$I_D \text{ (mA)}$
$V_{GS} \leq 3$	0
3	0
4	0.5
5	2.0
6	4.5
7	8.0
8	12.5
9	18.0

Plot these points and join smoothly to get the MOSFET transfer curve.

FINAL ANSWERS (Summary)

1) Gate voltage: $V_G = 9 \text{ V}$

2) Load line equation: $V_{GS} = 9 - 2 I_D \quad (I_D \text{ in mA})$

3) Load line intercepts: $(0 \text{ V}, 4.5 \text{ mA})$ and $(9 \text{ V}, 0 \text{ mA})$

4) MOSFET constant: $k = 0.5 \text{ mA/V}^2$

5) Device equation: $I_D = 0.5 (V_{GS} - 3)^2 \quad (\text{mA})$

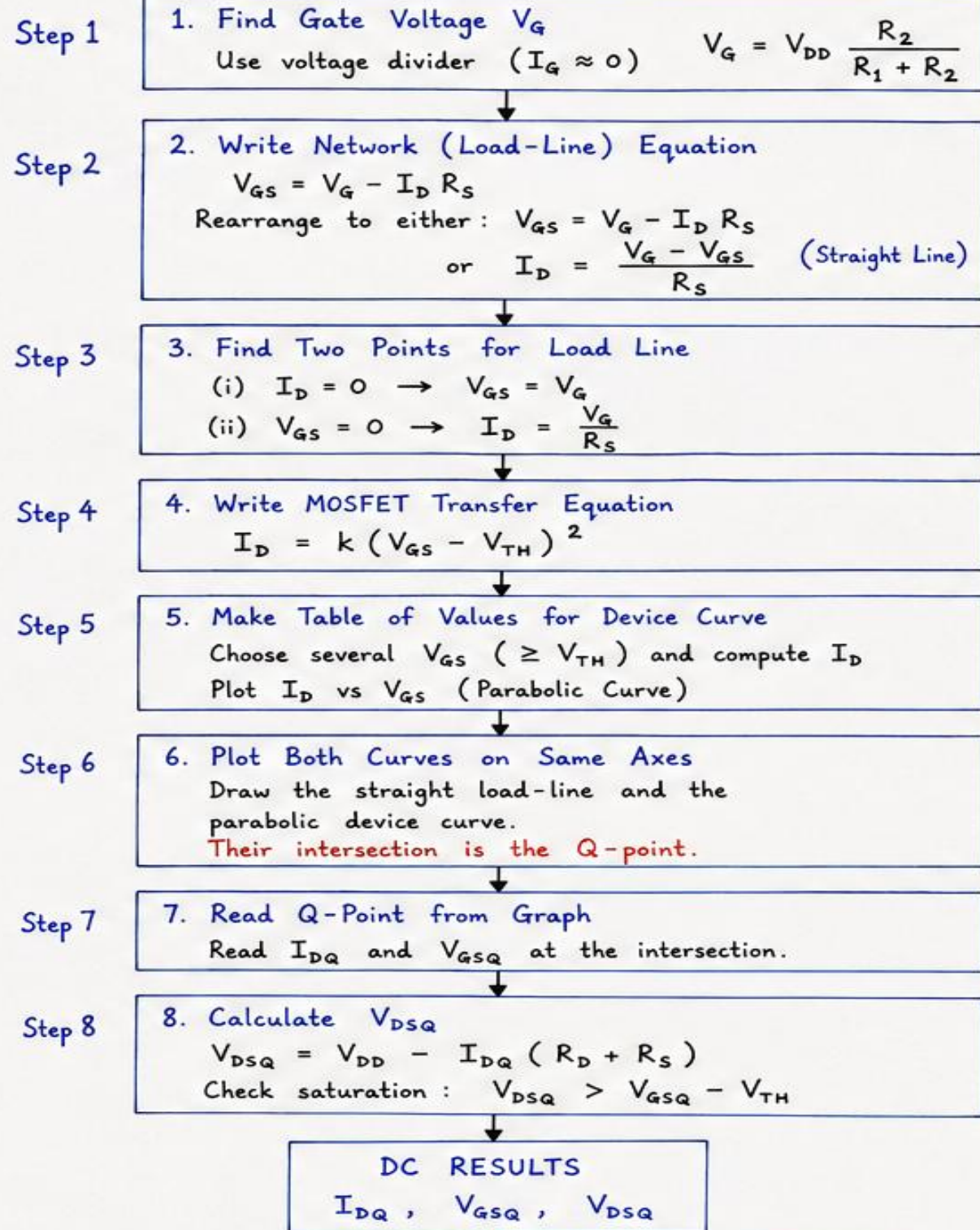
6) Q-point: $I_{DQ} = 2 \text{ mA}$, $V_{GSQ} = 5 \text{ V}$

7) Output voltage: $V_{DSQ} = 8 \text{ V}$

8) Saturation check: $8 \text{ V} > 2 \text{ V}$ ✓

E-MOSFET DC ANALYSIS METHODS – COMPARISON FLOW CHART

A) GRAPHICAL METHOD (using Graph)

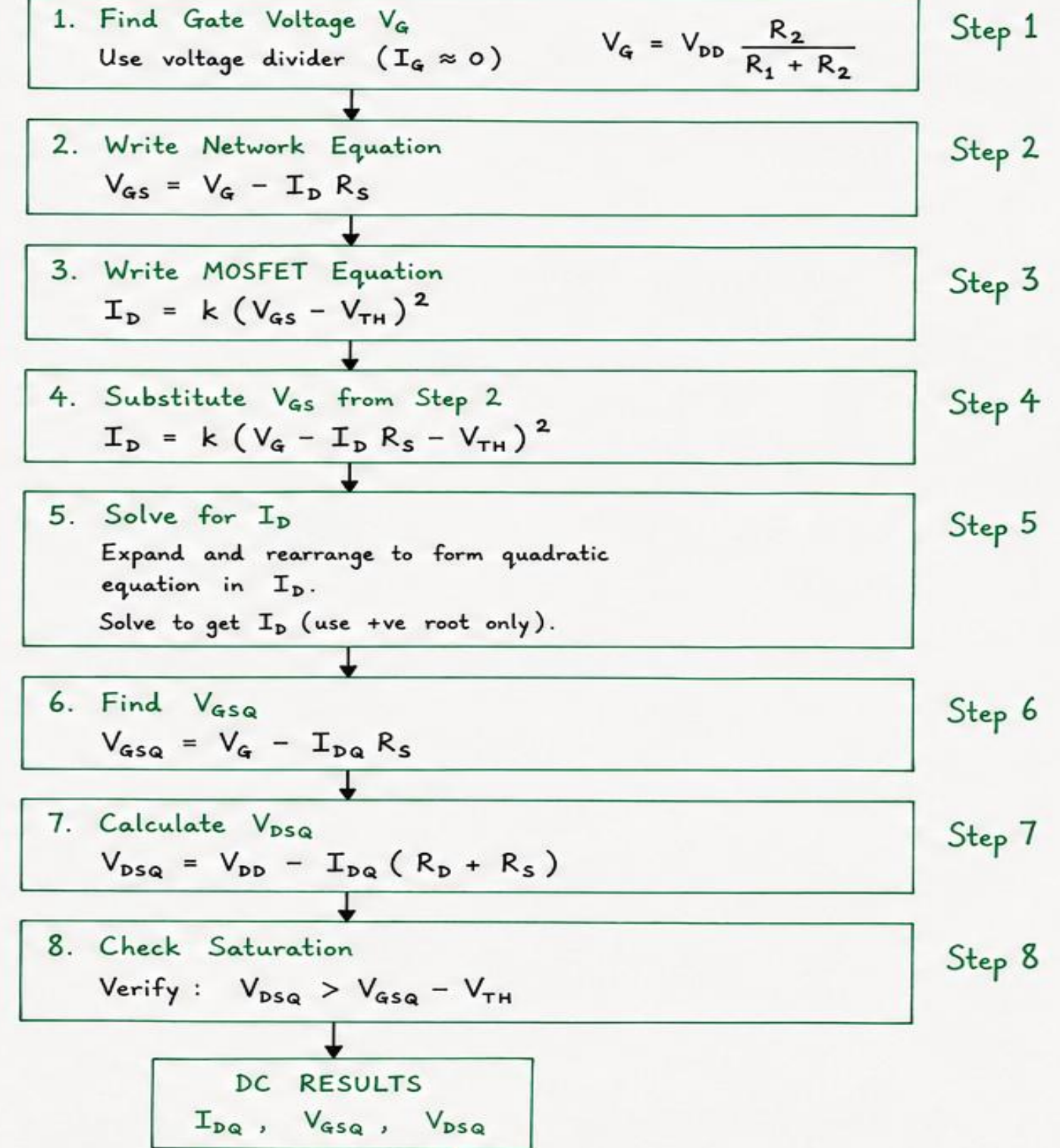


START

KEY DIFFERENCE

- Graphical Method
Uses plotting of network line and device curve.
Q-point is read from intersection.
 - Equation Method
Uses algebra to solve simultaneous equations and compute Q-point.
- Both methods give the same DC results.

B) EQUATION METHOD (using Algebra)

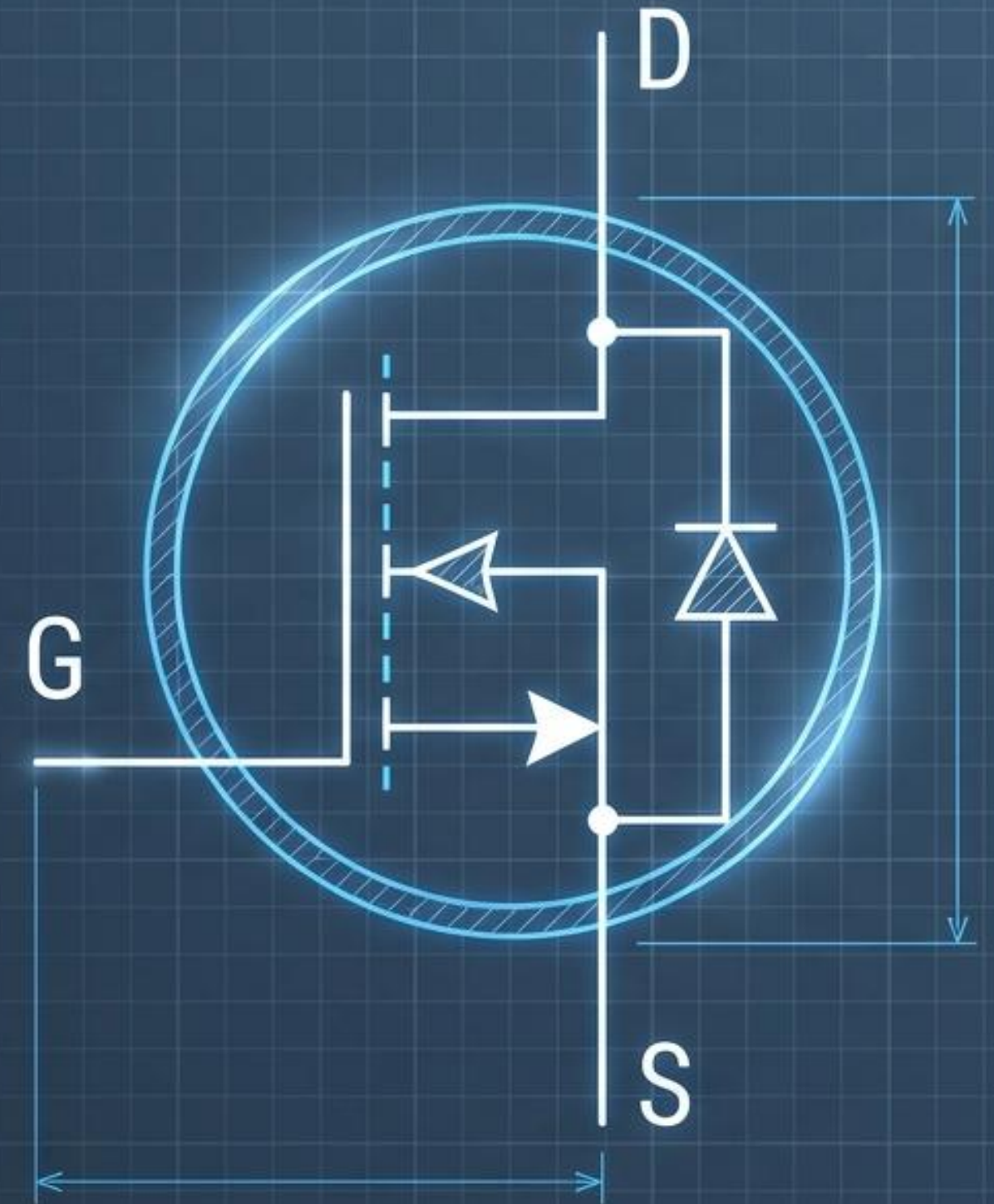


FINAL DC OUTPUT (BOTH METHODS)

I_{DQ} (Drain Current), V_{GSQ} (Gate-Source Voltage), V_{DSQ} (Drain-Source Voltage)
(Ensure MOSFET operates in saturation region.)

E-MOSFET Amplifiers in AC Systems

A visual primer on AC analysis, modeling,
and application exclusively for
Enhancement-Type Metal-Oxide-Semiconductor
Field-Effect Transistors.

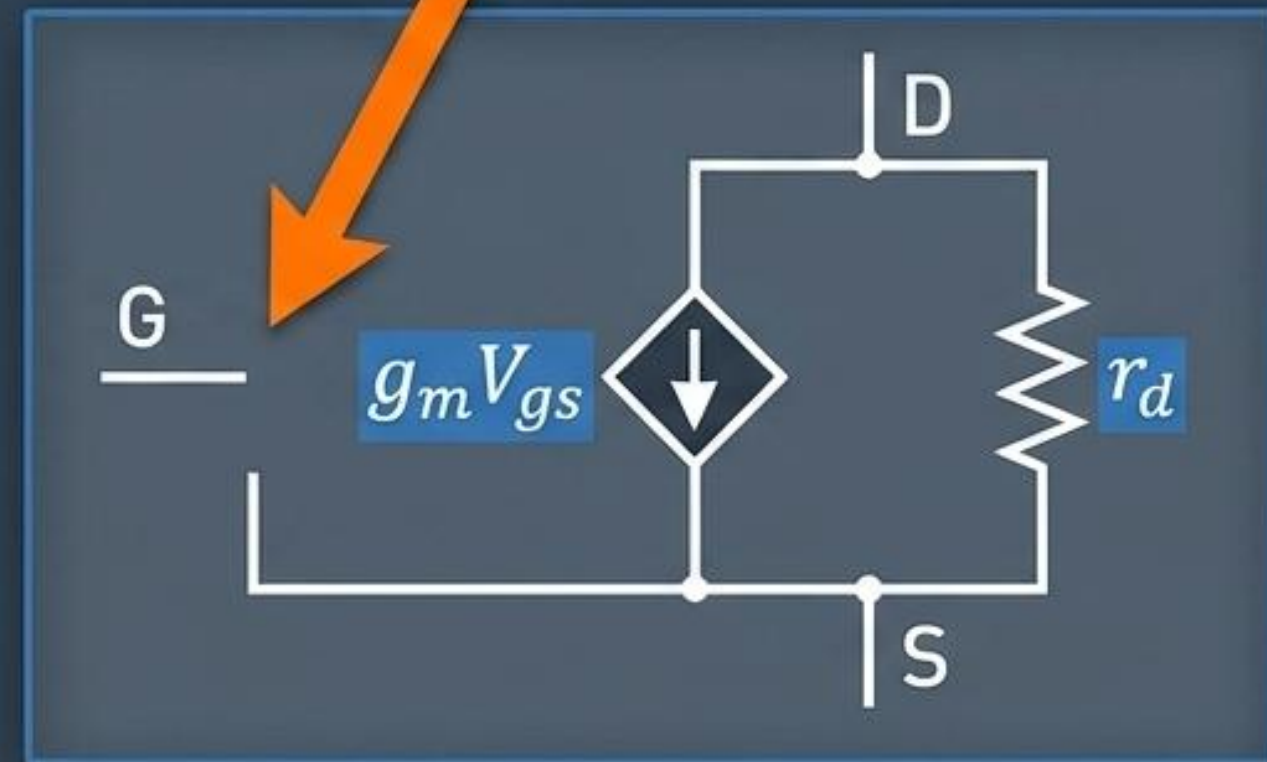


Modeling the E-MOSFET for AC analysis



Translation Panel

Open-Circuit between gate and drain-source channel.



Engineering Note

Values for g_m and r_d are extracted directly from the specific FET specification sheet.

Transconductance dictates the input-to-output relationship

Constant k is determined from the spec sheet's typical operating point.

$$I_D = k(V_{GS} - V_{GS(Th)})^2$$

The Derivative Action

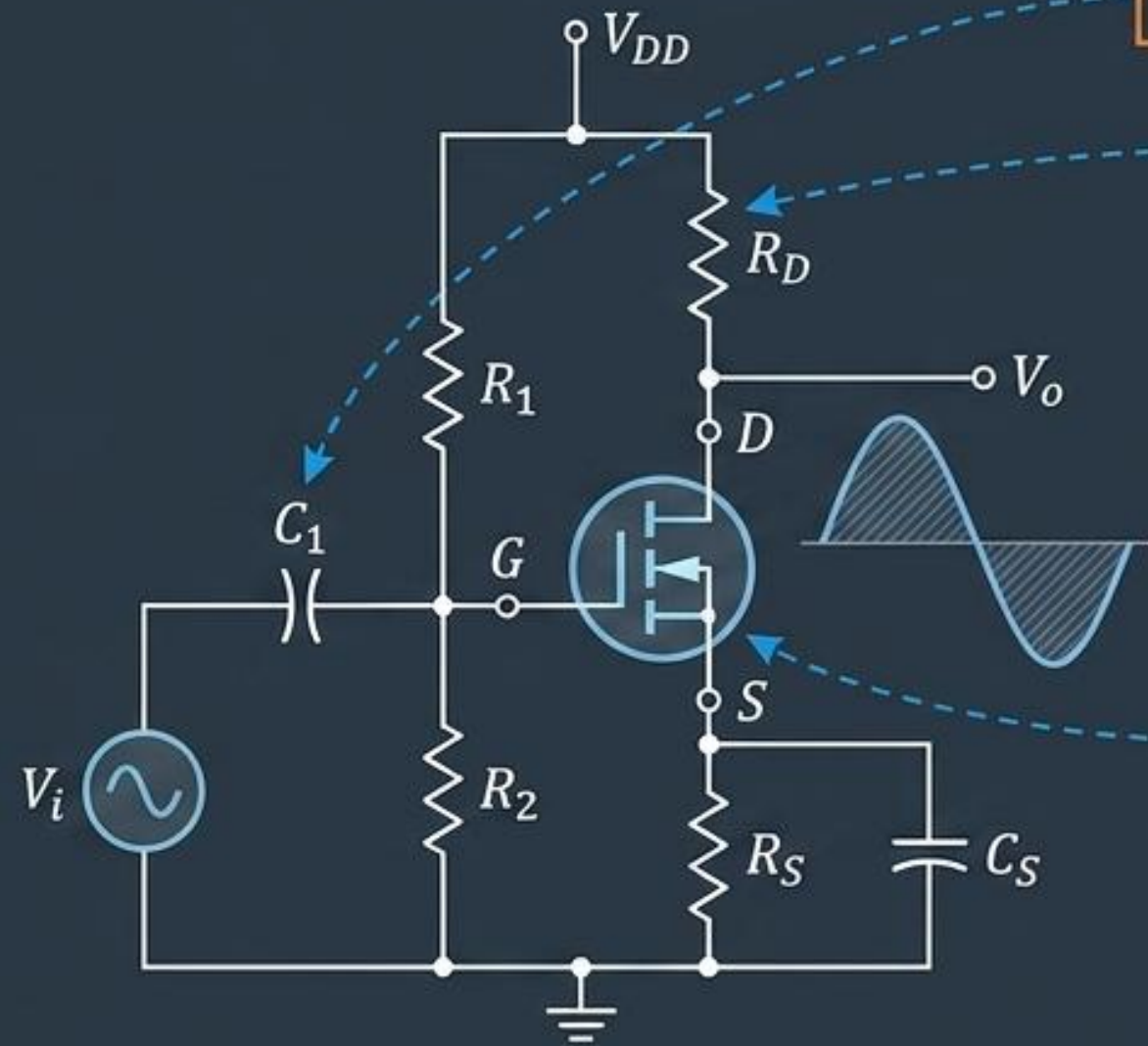
$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{d}{dV_{GS}} k(V_{GS} - V_{GS(Th)})^2$$

$$g_m = 2k(V_{GS_Q} - V_{GS(Th)})$$

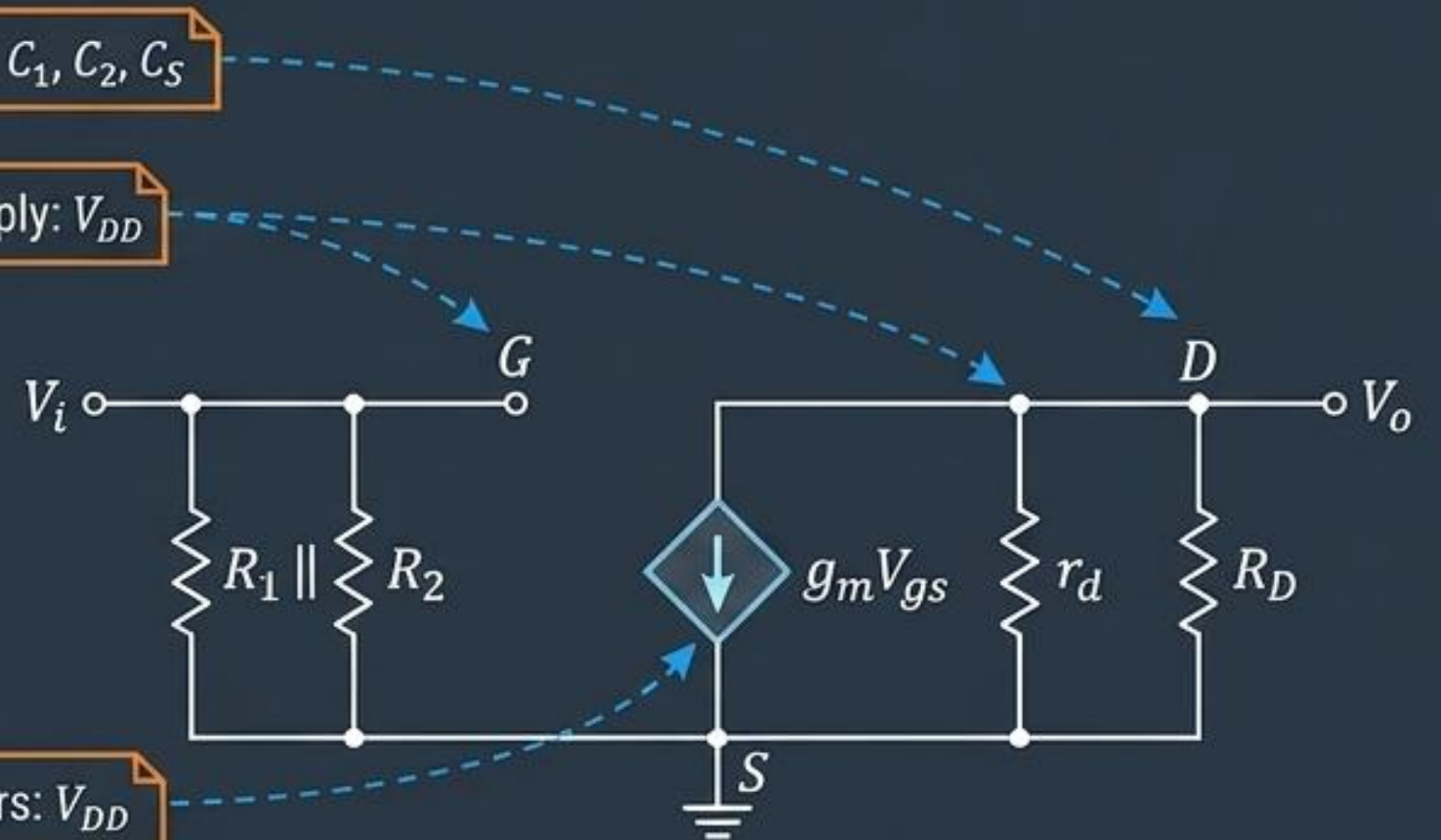
[This bridges the DC operating point to the AC analysis.]

Transitioning to the AC equivalent schematic

Physical Circuit Schematic



AC Equivalent Schematic



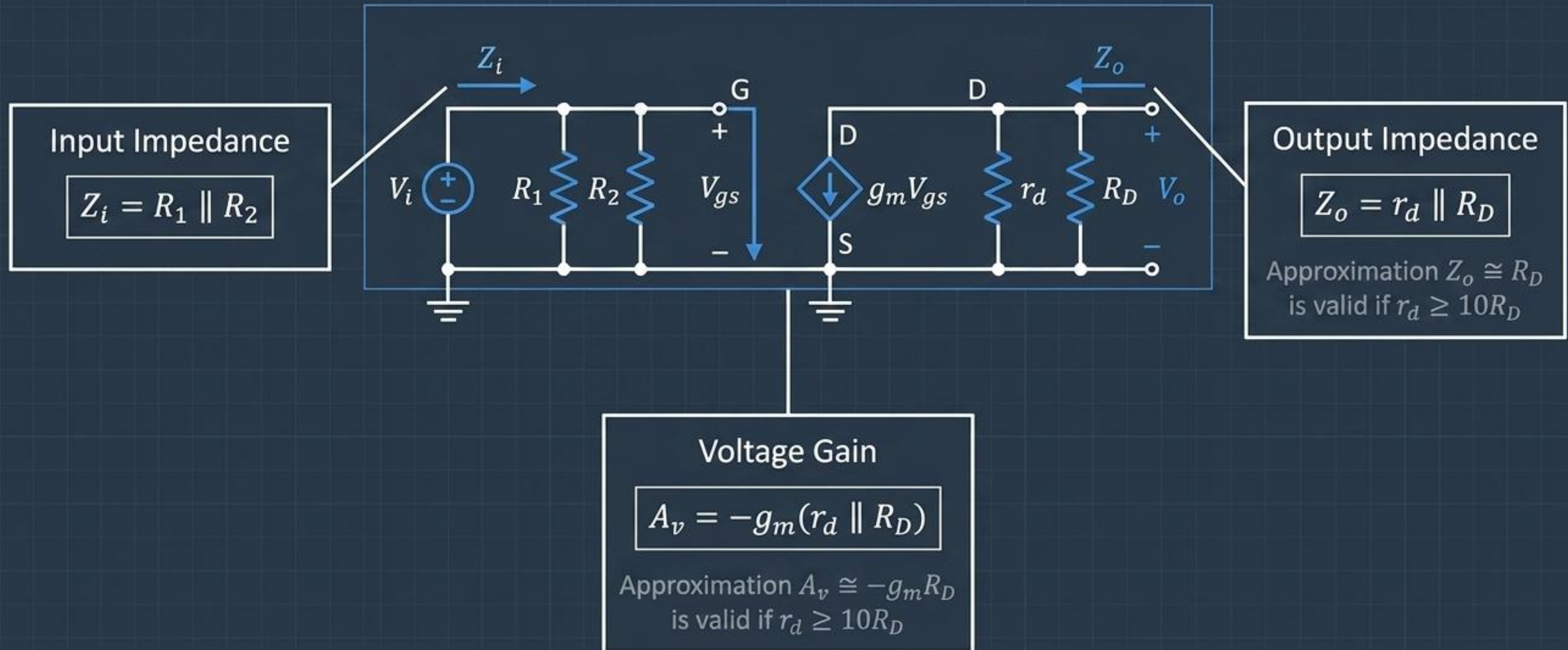
Short Capacitors: C_1, C_2, C_S

Ground DC Supply: V_{DD}

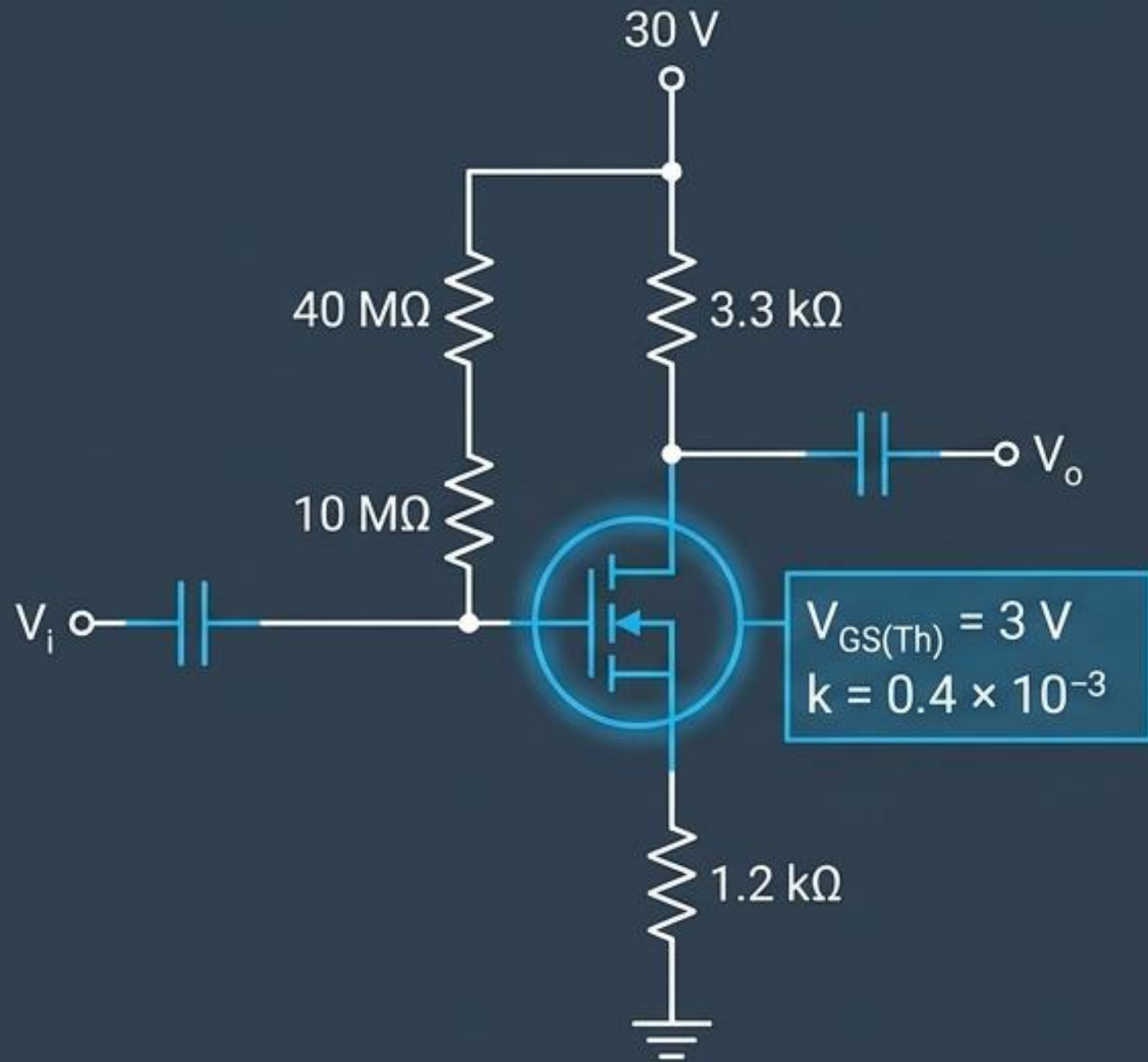
Short Capacitors: V_{DD}

180° Voltage Phase Shift
between Input and Output.

Governing equations map directly to the AC schematic



Defining the parameters for a practical amplifier



Known Variables

Input Signal (V_i):	0.8 mV
Internal Resistance (r_d):	40 k Ω
Threshold Voltage ($V_{GS(Th)}$):	3 V
Device Constant (k):	0.4×10^{-3}
Operating Point (V_{GS_Q}):	4.8 V (Pre-calculated for this example)

Objective:
Determine Output Voltage (V_o)

Executing the AC analysis yields the final output

Execution Flowchart

Step 1: Determine Transconductance

$$g_m = 2k(V_{GSQ} - V_{GS(Th)})$$

$$2(0.4 \times 10^{-3})(4.8 \text{ V} - 3 \text{ V})$$

$$g_m = 1.44 \text{ mS}$$

Step 2: Calculate System Gain

$$A_v = -g_m(r_d \parallel R_D)$$

$$-(1.44 \text{ mS})(40 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega)$$

$$A_v = -4.39$$

Step 3: Resolve Output Voltage

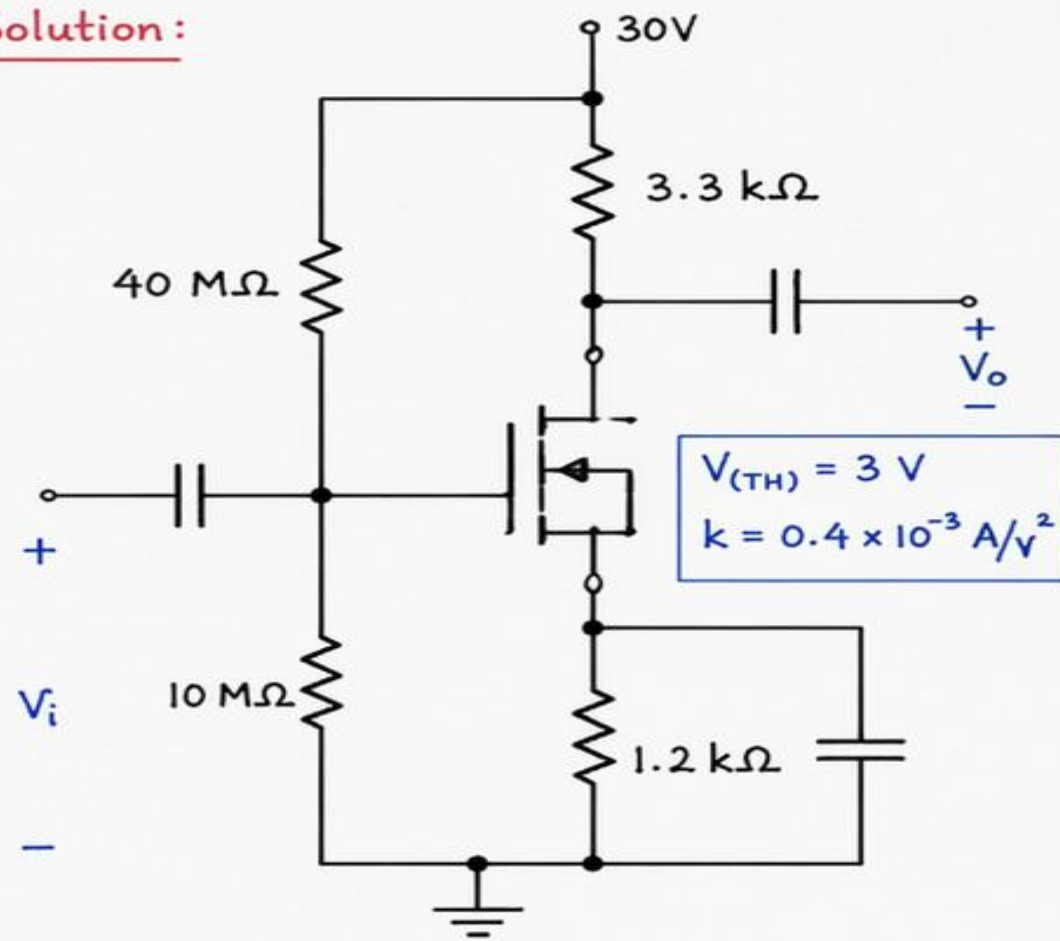
$$V_o = A_v V_i$$

$$(-4.39)(0.8 \text{ mV})$$

The negative sign mathematically confirms the 180° phase shift.

$$\mathbf{-3.51 \text{ mV}}$$

Solution:



Given: $r_d = 40 \text{ k}\Omega$, $v_i = 0.8 \text{ mV}$

1) Find DC bias (I_D , V_{GS})

Gate voltage from voltage divider:

$$V_G = \frac{10}{40+10} \times 30 = 6 \text{ V}$$

$$V_G = 6 \text{ V}$$

For DC (capacitors are open)

$$V_{GS} = V_G - I_D R_S = 6 - 1.2 \text{ k} I_D$$

In saturation: $I_D = k (V_{GS} - V_{TH})^2$

$$I_D = 0.4 \times 10^{-3} (6 - 1.2 \text{ k} I_D - 3)^2$$

$$= 0.4 \times 10^{-3} (3 - 1.2 \text{ k} I_D)^2$$

Solving: $I_D = 5.625 \text{ mA}$ (rejected)

$$I_D = 1.111 \text{ mA}$$

$$V_{GS} = 6 - 1.2 \text{ k} (1.111 \text{ mA})$$

$$= 6 - 1.333 \approx 4.67 \text{ V}$$

$$V_{GS} \approx 4.67 \text{ V}$$

Check saturation:

$$V_{DS} = 30 - I_D (3.3 \text{ k}) - I_D (1.2 \text{ k})$$

$$= 30 - 1.111 \text{ mA} (4.5 \text{ k})$$

$$= 25.0 \text{ V}$$

$$V_{GS} - V_{TH} = 4.67 - 3 = 1.67 \text{ V}$$

$$V_{DS} (25 \text{ V}) > V_{GS} - V_{TH} (1.67 \text{ V})$$

$\therefore$ Saturation condition satisfied

2) Small-signal transconductance

$$g_m = 2 k (V_{GS} - V_{TH})$$

$$= 2 (0.4 \times 10^{-3}) (4.67 - 3)$$

$$= 1.33 \times 10^{-3} \text{ S}$$

$$g_m \approx 1.33 \text{ mS}$$

3) Small-signal voltage gain

$$R_D \parallel r_d = \frac{3.3 \text{ k} \times 40 \text{ k}}{3.3 \text{ k} + 40 \text{ k}}$$

$$= \frac{132 \text{ k}}{43.3} \approx 3.05 \text{ k}\Omega$$

$$A_v = -g_m (R_D \parallel r_d)$$

$$= - (1.33 \times 10^{-3}) (3.05 \times 10^3)$$

$$A_v \approx -4.05$$

4) Output voltage

$$v_o = A_v v_i$$

$$= (-4.05) (0.8 \text{ mV})$$

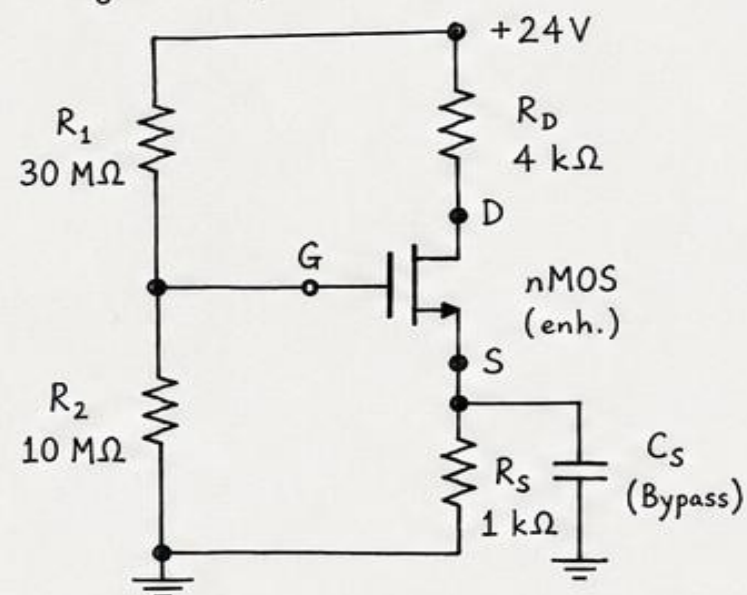
$$v_o \approx -3.24 \text{ mV}$$

Final Answers: Voltage gain: $A_v \approx -4.05$

Output voltage: $v_o \approx -3.24 \text{ mV}$

EXAMPLE : E-MOSFET BIASING (Graphical) AND AC ANALYSIS

For the circuit shown, determine the DC operating point graphically and find the small-signal voltage gain and output voltage for $v_i = 1 \text{ mV}$.



REQUIRED :

- 1) Find gate voltage V_G .
- 2) Write the network (load-line) equation relating I_D and V_{GS} .
- 3) Find two points to draw the load line.
- 4) Write the MOSFET transfer equation and make a table.
- 5) Plot the load line and transfer curve and find Q-point.
- 6) Calculate V_{DSQ} and verify saturation.
- 7) Find g_m .
- 8) Find $R_D' = R_D \parallel r_d$.
- 9) Find voltage gain A_v .
- 10) Find output voltage v_o .

6) Drain-source voltage V_{DSQ}

$$\begin{aligned} V_{DS} &= V_{DD} - I_D (R_D + R_S) \\ &= 24 - (2 \text{ mA})(4 \text{ k}\Omega + 1 \text{ k}\Omega) \\ &= 24 - (2 \text{ mA})(5 \text{ k}\Omega) \\ &= 24 - 10 \end{aligned}$$

$$\boxed{V_{DSQ} = 14 \text{ V}}$$

Check saturation :

$$V_{GS} - V_{TH} = 4 - 2 = 2 \text{ V}$$

$$V_{DS} = 14 \text{ V} > 2 \text{ V} \rightarrow \text{Saturation } \checkmark$$

7) Small-signal transconductance g_m

$$\begin{aligned} g_m &= 2k (V_{GS} - V_{TH}) \\ &= 2 (0.5 \times 10^{-3}) (4 - 2) \\ &= 2 \times 10^{-3} \text{ S} \end{aligned}$$

$$\boxed{g_m = 2 \text{ mS}}$$

$$(\text{Check using } I_D : g_m = 2 I_D / V_{ov})$$

$$V_{ov} = V_{GS} - V_{TH} = 2 \text{ V}$$

$$g_m = 2 (2 \text{ mA}) / 2 \text{ V} = 2 \text{ mS } \checkmark$$

1) Gate voltage (V_G)

Since $I_G \approx 0$, the gate voltage is given by the divider.

$$\begin{aligned} V_G &= V_{DD} \frac{R_2}{R_1 + R_2} \\ &= 24 \times \frac{10}{30 + 10} = 24 \times \frac{1}{4} \end{aligned}$$

$$\boxed{V_G = 6 \text{ V}}$$

2) Network equation (Load line)

$$\begin{aligned} V_{GS} &= V_G - V_S = V_G - I_D R_S \\ &= 6 - I_D (1 \text{ k}\Omega) \end{aligned}$$

$$\boxed{V_{GS} = 6 - I_D \quad (I_D \text{ in mA})}$$

Equivalent form for graph :

$$\boxed{I_D = 6 - V_{GS} \quad (I_D \text{ in mA})}$$

3) Two points for load line

$$(i) I_D = 0 \rightarrow V_{GS} = 6 - 0 = 6 \text{ V}$$

$$\boxed{\text{Point A : } (V_{GS}, I_D) = (6 \text{ V}, 0 \text{ mA})}$$

$$(ii) V_{GS} = 0 \rightarrow I_D = 6 - 0 = 6 \text{ mA}$$

$$\boxed{\text{Point B : } (V_{GS}, I_D) = (0 \text{ V}, 6 \text{ mA})}$$

Draw straight line joining A and B. This is the **NETWORK LOAD LINE**.

4) MOSFET transfer equation

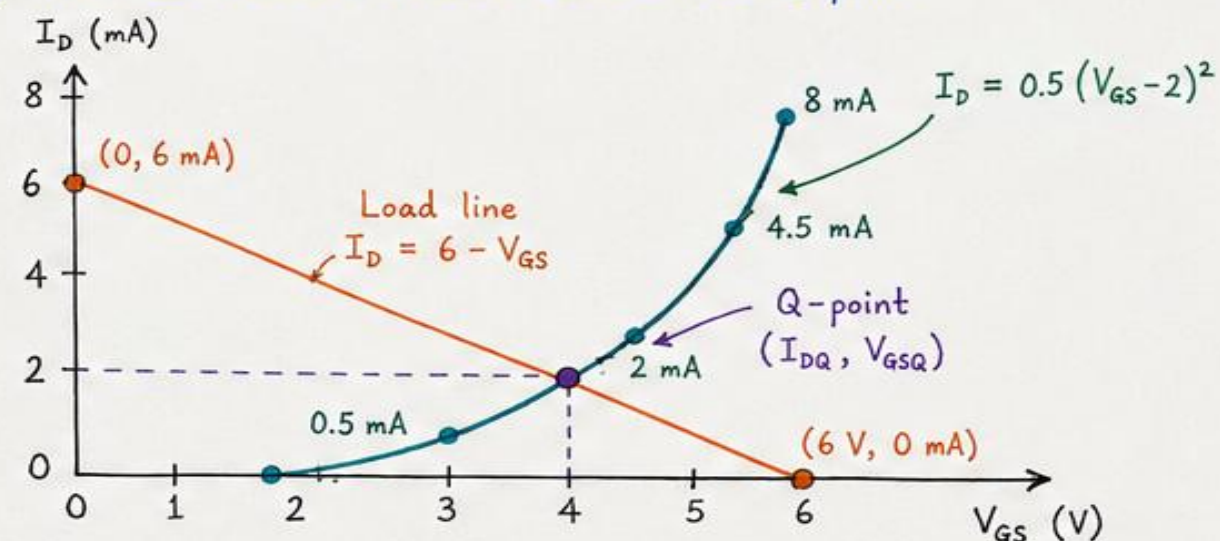
$$\begin{aligned} I_D &= k (V_{GS} - V_{TH})^2 \\ &= 0.5 (V_{GS} - 2)^2 \quad (I_D \text{ in mA}) \end{aligned}$$

Make a table :

$V_{GS} \text{ (V)}$	2	3	4	5	6
$I_D \text{ (mA)}$	0	0.5	2.0	4.5	8.0

(For $V_{GS} \leq 2 \text{ V}$, $I_D \approx 0$)

5) Plot load line and transfer curve - find Q-point



From the graph : $I_{DQ} \approx 2 \text{ mA}$, $V_{GSQ} \approx 4 \text{ V}$

8) AC drain resistance

$$\begin{aligned} R_D' &= R_D \parallel r_d = 4 \text{ k}\Omega \parallel 40 \text{ k}\Omega \\ &= \frac{4 \text{ k} \times 40 \text{ k}}{4 \text{ k} + 40 \text{ k}} \\ &= \frac{160 \text{ k}}{44} \approx 3.64 \text{ k}\Omega \end{aligned}$$

$$\boxed{R_D' \approx 3.64 \text{ k}\Omega}$$

9) Voltage gain A_v

(R_S is bypassed for AC $\rightarrow$ source $\approx$ ac ground)

$$\begin{aligned} A_v &= -g_m (R_D \parallel r_d) \\ &= - (2 \times 10^{-3}) (3.64 \times 10^3) \\ &= -7.28 \end{aligned}$$

$$\boxed{A_v \approx -7.28}$$

Negative sign $\rightarrow$ output is 180° out of phase with input.

10) Output voltage v_o

$$\begin{aligned} v_o &= A_v v_i \\ &= (-7.28) (1 \text{ mV}) \end{aligned}$$

$$\boxed{v_o \approx -7.28 \text{ mV}}$$

FINAL ANSWERS

$$\begin{aligned} I_{DQ} &\approx 2 \text{ mA} & g_m &\approx 2 \text{ mS} \\ V_{GSQ} &\approx 4 \text{ V} & A_v &\approx -7.28 \\ V_{DSQ} &\approx 14 \text{ V} & v_o &\approx -7.28 \text{ mV} \end{aligned}$$

Saturation condition satisfied. $\checkmark$